EXPLORING FACTORS INFLUENCING FATAL TRAFFIC CRASHES IN THE UNITED STATES (2015-2022): SPATIAL AND NON-SPATIAL CLUSTERING APPROACH

Project Report (DSA 5099)

Submitted by

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A project report submitted to the Department of Mathematical and Physical Sciences, East West University, in partial fulfilment of the requirement for the degree of Master of Science in Data Science and Analytics

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DEDICATION

To my precious newborn daughter, whose arrival has brought immeasurable joy and love into my life.

ABSTRACT

This study examines patterns in fatal traffic crashes in the United States based on spatial and non-spatial clustering approach. The analysis uses FARS (Fatality Analysis Reporting System) data from the National Highway Traffic Safety Administration (NHTSA) for 2015–2022. The person-level dataset contains all 308,704 people who died in 284,193 fatal crashes in the 50 states and the District of Columbia. The study utilizes Optimized Hot Spot Analysis and K-mode clustering to examine the spatial distribution of fatal crashes and non-spatial analysis to identify key patterns among demographic factors (Age range, Sex, Race) and road-related factors. In addition to this, a Chi-square test of independence was applied to assess the relationship between road hierarchy and other categorical variables such as road-related factors and weather type based on road type.

The spatial analysis is conducted with the help of ArcGIS Pro which identified concentrations of fatal crashes on different road types. The largest concentrations were found on Arterial and Local roads in regions such as the Northeast and Southern regions of the U.S. Because the maps are based on counts, they partly reflect where people live and drive; rates per vehicle-mile are needed before these regions can be called higher-risk. Non spatial clustering approach like K-mode clustering grouped the fatalities into three descriptive profiles: daytime intersection crashes with older victims (45%), night-time weekend run-off-road crashes on unlit two-lane roads (33%), and crashes on high-speed divided highways and interstates (23%). The clusters are weakly separated (silhouette 0.10), so they are reported as descriptive profiles. With more than 280,000 crashes every chi-square test was significant; the effect sizes show that road class is associated mainly with road design (speed limit, Cramér's $V = 0.37$; divided or undivided trafficway, $V = 0.34$) and only negligibly with the characteristics of the person who died ($V \leq 0.07$).

Because FARS contains only fatal crashes and no exposure data, shares such as the 72% of victims recorded as White or the 71% of deaths in clear weather describe fatal crashes rather than risk. The result of this research highlight the complex relationship of road infrastructure, weather conditions, and demographic factors in fatal crashes. The integration of spatial hot spot analysis, clustering, and statistical tests gives a reproducible description of where and under which conditions fatal crashes occur. These can be helpful for the development of focused interventions to improve road safety across the United States.

ACKNOWLEDGEMENTS

I would like to express my gratitude to Almighty Allah and give sincere thanks to my project supervisor, Dr. Md. Rezaul Karim, for his invaluable guidance throughout my research. He dedicated a lot of his precious time to help me exploring various topics. His personal interest in my work is a great source of motivation and inspiration. Finally, I would like to express my heartfelt thanks to all my teachers and friends for their constant support.

CERTIFICATION

This is to certify that this project was submitted to the Department of Mathematical and Physical Sciences and evaluated by the project examination committee. The committee accepted this project as a fulfillment of the requirements for the degree of Master of Science.

_____

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DECLARATION

I, Md. Rakibul Hasan Himal, hereby declare that this is a presentation of my original research work. Wherever contributions of others are involved, every effort is made to indicate this clearly, with due reference to literature.

This project has been composed by me and has not been submitted in any previous application for any degree or qualification at this or any other institution. The work reported herein has been carried out by me under the supervision of Dr. Md. Rezaul Karim at East West University.

I have duly referred to and acknowledged the work of others and all sources of information used in the preparation of this project. This project complies with the ethical standards and guidelines set forth by East West University.

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ETHICS STATEMENT

This research was conducted in strict accordance with the ethical standards and guidelines provided by East West University. All procedures performed in this study complied with the ethical standards of the institution

Data handling and analysis were conducted following the guidelines provided by East West University ensuring confidentiality and data security throughout the study. The study used only publicly available, de-identified secondary data from NHTSA's Fatality Analysis Reporting System; no human participants were recruited, so informed consent was not required.

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Contents

DEDICATION i
ABSTRACT ii
ACKNOWLEDGEMENTS iii
CERTIFICATION iv
DECLARATION v
ETHICS STATEMENT vi
LIST OF FIGURES ix
LIST OF TABLES x

CHAPTER 1:INTRODUCTION 1

1.1 Background of the Study 1
1.2 Contribution and Motivation of the Study 2
1.3 Research Gap and Research question 2
1.4 Objectives of the Study 3
1.4.1 Main Objectives 3
1.4.2 Specific Objectives 3
1.5 Limitation of the study 3
1.6 Research Outline 4

2 CHAPTER 2: LITERATURE REVIEW 5

2.1 Introduction 5
2.1.1 Road Infrastructure 5
2.1.2 Driver Behavior and Demographics 6
2.1.3 Environmental and Weather Factors 7
2.1.4 Advanced Analytical Techniques 7
2.2 Concluding remarks 8

3 CHAPTER 3: METHODOLOGY FOR THE RESEARCH 9

3.1 Introduction 9
3.2 Data Collection and Preprocessing 9
3.2.1 Data source and files 9
3.2.2 Building the person-level dataset 10
3.3 Data Cleaning and Transformation 15
3.3.1 Harmonising categories across years 15
3.3.2 Derived variables 16
3.3.3 Data quality 16
3.3.4 Crash locations for the spatial analysis 16
3.3.5 Analysis-ready datasets 17
3.4 Chi-square Test of Association 17
3.5 Spatial Analysis: Hot Spot Analysis Based on Road Hierarchy 18
3.6 Non-Spatial Analysis: K-Mode Clustering 19
3.7 Results and Interpretation 20

3.8 Analytical tools and software 21
3.9 Concluding remarks 21
4 CHAPTER 4: DATA ANALYSIS AND RESULTS 22
4.1 Introduction 22
4.2 Exploratory data analysis 22
4.2.1 Geographical Context 22
4.2.2 State-wise Yearly Percentage Change in Fatalities (Year-over-Year) 23
4.2.3 Seasonal Fatal crashes Trends (2015-2022) 24
4.2.4 Fatal Crash Count Based on Weekday and Weekend 26
4.2.5 Fatal Crash Count Based on Road Hierarchy 26
4.2.6 Top Ten and Bottom Ten Fatal Crash rates per 100,000 Population 28
4.2.7 Weather, Light, and Surface Condition-Based Fatal Accident Distribution 29
4.2.8 Age, Race, and Sex-Based Fatal Accident Distribution 30
4.3 Spatial and Non Spatial cluster Analysis 31
4.3.1 Testing the Independent Association of road hierarchy 31
4.3.2 Spatial Clustering (Hot Spot analysis) 33
4.3.3 Non Spatial Clustering: K-Mode Clustering 37
4.3.4 Cluster Characteristics 37
4.4 Concluding remarks 41
5 CHAPTER 5: DISCUSSION 42
5.1 Introduction 42
5.2 Summary of the Main Findings 42
5.3 Comparison with Previous Studies 43
5.4 What the Methods Can and Cannot Show 44
5.5 Implications for Road Safety Practice 45
5.5.1 Night-time Run-off-road Crashes on Two-lane Roads 45
5.5.2 Crashes on High-speed Divided Highways 45
5.5.3 Daytime Intersection Crashes and Older Road Users 45
5.5.4 Pedestrians on Arterial Roads at Night 46
5.5.5 Targeting by Location 46
5.6 Limitations 46
5.7 Concluding Remarks 46
6 CHAPTER 6: CONCLUSION AND FURTHER RESEARCH 47
6.1 Conclusion 47
6.2 Contribution of the Study 48
6.3 Area of Further Research 48
7 REFERENCES 49
8 APPENDIX 58

LIST OF FIGURES

Figures **Page**

Figure 3.1 Workflow from the FARS Files to the Analysis-ready Datasets 12
Figure 4.1 USA State Map 22
Figure 4.2 State-wise Yearly Percentage Change in Fatalities (Year-over-Year) 23
Figure 4.3 Average Fatalities per Day by Season (2015-2022) 25
Figure 4.4 Crash Count Based on Weekday and Weekend 27
Figure 4.5 Crash Count Based on Road Hierarchy 27
Figure 4.6 Crash location Hot Spot analysis based on Arterial Road 35
Figure 4.7 Crash location Hot Spot analysis based on Collector Road 35
Figure 4.8 Crash location Hot Spot analysis based on Interstate Road 36
Figure 4.9 Crash location Hot Spot analysis based on Local road 36
Figure 4.10 K-modes Diagnostics for Choosing k (Cost, Silhouette and Seed Stability) 37
Figure 4.11 K-modes Clusters on the First Two MCA Dimensions 40

LIST OF TABLES

Tables **Page**

Table 3.1 List of Variables and levels 13
Table 3.2 Fatal Crashes and Fatalities by Year, 2015–2022 14
Table 3.3 Analysis-ready Datasets 17
Table 4.1 Top Ten and Bottom Ten Fatal Crash rates per 100,000 Population 28
Table 4.2 Weather, Light and Surface Condition-Based Fatal Accident Distribution 29
Table 4.3 Age, Race, and Sex-Based Fatal Accident Distribution 30
Table 4.4 Chi Square test results 32
Table 4.5 Cluster Table 38

Chapter 1
Introduction

1.1 Background of the Study

Traffic accidents are a serious problem for public health and safety around the world. In the United States, many people die, get injured, and face economic losses because of road accidents. According to the National Highway Traffic Safety Administration (NHTSA), more than 42,000 people died in traffic accidents in 2021, and millions were injured (NHTSA, 2022). These numbers show why it is important to find and reduce the causes of road accidents.

Older studies often looked at single factors like driver behavior, road conditions, or the environment (Cicchino and McCartt, 2015). But a broader approach using spatial analysis and machine learning can give deeper understanding and better predictions.

Spatial analysis looks at where accidents happen and helps find high-risk locations and times. Studies show that most accidents occur in cities and change depending on the time of day, weather, and seasons.When combined with machine learning, spatial data can reveal complex patterns that are hard to see otherwise. Machine learning methods like for clustering variables decision trees and support vector machines and for prediction neural networks, help analyze large datasets and make better predictions about accidents.

Though there are significant improvements in traffic safety research, it is still difficult to understand how different factors affect accidents. Individual characteristics, traffic condition, road quality, weather interact with each other in complicated ways. This study will use spatial and non-spatial clustering to fill knowledge gaps This will help build better policies to reduce traffic accidents in the United States.

1.2 Contribution and Motivation of the Study

This research helps to understand the factors behind fatal traffic accidents in the United States. It looks at both location-based (spatial) and non-location-based (non-spatial) patterns. The study uses different methods, including spatial analysis (Optimized Hot Spot Analysis), non-spatial clustering (K-mode clustering), and statistical analysis (Chi-square test). These methods help find high-risk areas and important factors like age, sex, weather, and road types that contribute to fatal accidents. This study identifies accident hot spots on different types of roads. It also groups important factors that affect fatal crashes. The research is useful for making better safety measures, public policies, and road improvements. It focuses on high-risk areas and protects road users.

Traffic fatalities represent a significant public safety concern for the whole world. By analyzing the factors contributing to fatal crashes is essential for reducing accident which can lead to save lives. In spite of efforts to improve road safety, fatal crash rates remain a major issue. In USA, certain regions and road types demonstrated repeatedly high accident frequencies. By identifying these high-risk zones and understanding the underlying factors can help policymakers, urban planners, and transportation authorities to implement more specific safety measures. This research provides valuable insights into these issues. This can lead to more effective safety interventions and strategies.

1.3 Research Gap and Research question

Various type of studies have examined traffic accidents and fatalities, but the usage of the integration of spatial and non-spatial clustering methods to analyze the spatial distribution and underlying patterns in crash data remains under-explored. Furthermore, most of the previous research has focused on different distinct factors, such as road type or weather condition. This research connect this gap by using K-mode clustering and Hot Spot Analysis to capture

complex interactions among road types, weather conditions, demographic factors, and crash outcomes across a large dataset spans across eight years. So the primary research question will be:

What are the group wise key characteristics associated with fatal traffic crashes in the United States, and how are the states spatially connected based on the road hierarchy?

1.4 Objectives of the Study

1.4.1 Main Objectives

To identify spatial patterns of fatal traffic crashes across different road hierarchies and categorize the factors related to the occurrence of fatal crashes in the United States

1.4.2 Specific Objectives

The specific objectives of this research are:

I. to conduct a spatial hot spot analysis of fatal crashes to identify areas with a high concentration of traffic fatalities in the U.S. based on different road types;

II. to apply K-mode clustering to group fatalities by their crash, road and person characteristics to identify patterns in crash characteristics;

III. to perform a Chi-square test to assess the statistical significance of the relationships between road hierarchy and other categorical variables;

IV. to develop policy recommendations and safety interventions defined for the identified crash patterns and high-risk areas based on the findings of the spatial and non-spatial analyses.

1.5 Limitation of the study

The analysis was conducted at the state level which can hide the local variation in crash patterns. With local level analysis, it can provide localized risk factors. This study only included fatal crash data excluding the number of injured people which could provide a broader

understanding of overall road safety dynamics. Many continuous variables were converted into categorical variables which might lead to loss of variability. FARS contains only fatal crashes and no exposure data (vehicle-miles travelled, population at risk), so the study describes the characteristics of fatal crashes and cannot estimate risk or the effect of a factor on the chance of a crash being fatal. Several variables are police-reported and partly missing (for example, pavement type is unknown for 25% of fatalities and race for 6%; race is redacted for Iowa from 2020). Person-level summaries count a crash once for each person who died in it. The hot spot analysis uses crash counts and is not normalised for traffic volume.

1.6 Research Outline

This research work is organized into six chapters:

Chapter 1: This chapter gives a brief discussion about background of the study, contribution and motivation of the study, research gap and question and objectives of the study.

Chapter 2: This chapter gives previous studies on traffic accidents and fatalities, review of spatial and non-spatial clustering techniques in crash analysis, how road infrastructure, weather and demographic factors affect fatal crashes.

Chapter 3: This chapter discusses in detail about the methodology of the study which is how the data is collected, how it is cleaned, how it is used in exploratory data analysis and further analysis

Chapter 4: This chapter contains exploratory data analysis before conducting spatial and non-spatial cluster analysis. After that it discusses the analysis and results of using the spatial and non-spatial clustering method.

Chapter 5: This chapter discusses the findings, compares them with previous studies, explains what the methods can and cannot show, and sets out the implications for road safety practice.

Chapter 6: The final chapter contains the discussion about further study and conclusion of the study.

Chapter 2
Literature Review

2.1 Introduction

Traffic accidents remain an important public safety issue in the United States influenced by a wide range of factors which driver behavior, road conditions, environmental influences, and technological advancements. Here we discuss about the previous studies.

2.1.1 Road Infrastructure

Road infrastructure elements significantly contribute to fatal traffic accidents across the United States. Multiple studies highlighted the critical role of design deficiency in severity and causation of crashes (Papadimitriou et al., 2019). There has been relationship between pavement surface conditions and traffic safety which reveals that worsening condition of road surfaces substantially increase crash risk. Besides pavement roughness measured by International Roughness Index (IRI) directly correlates with accident frequency, mostly in adverse weather condition where visibility is not good. (Abdel-Rahim & Ellsworth, 2021). Fifty nine specific risk factors have been identified including alignment deficiencies, cross-section characteristics, road surface deficiencies, and junction design problems by systematically analyzing Infrastructure-related crash risk factors. By properly addressing these issues, up to 44% crash rate can be reduced. (Papadimitriou et al., 2019). Federal Highway

Administration's research shows that geometric design features such as insufficient clear zones, inadequate shoulder widths, and poor sight distances at intersections contribute significantly to fatal outcomes (Federal Highway Administration, 2021). Moreover, absence of protective roadside infrastructure elements like continuous guardrails on expressways directly associated with increased fatality rates in roadway departure crashes (O'Neil & Marsh, 2019; Smart Growth America, 2022). Continious road safety audits as well as geometric design improvements can play a pivotal role to provide substantial safety benefits when applied systematically across road networks (National Highway Traffic Safety Administration, 2022; Wilson & Lipinski, 2000).

2.1.2 Driver Behavior and Demographics

Research by Cicchino and McCartt (2015) found that older drivers often make serious mistakes at intersections. Furthermore, older drivers, particularly those aged 75 and above, were more likely to be involved in fatal intersection crashes. The risk of fatal crashes at intersections significantly increased with age, particularly for drivers aged 85 and older (Preusser et al., 1997). Researchers have used experiments to discover what kinds of mistakes older drivers might make that could cause accidents. Drivers 65 and older were more likely than younger drivers to conclude that it was safe to pass through a junction after a danger entered the scene that rendered proceeding unsafe when they looked at pictures of traffic scenes for a short time (Caird et al., 2005). The results of simulator studies showed that older drivers reduced the size of the gap in traffic that they would accept as vehicle speeds increased more than younger drivers did. This provides evidence that older drivers pay more attention to the distance between themselves and another vehicle than the speed of the other vehicle when making decisions regarding gap acceptance (Staplin, 1995; Yan et al., 2007). Gender also plays a role in crash severity for older drivers. Cicchino and McCartt (2015) highlighted that older female drivers had a higher chance of both inadequate surveillance and misjudgment errors compared

to their male counterparts. A study by McCartt and Teoh (2015) showed that crash rates for teenagers dropped after the introduction of graduated driver licensing (GDL) programs. However, teenage drivers still take many risks. They often speed and drive under the influence. These risky behaviors are more common in new drivers who have little experience. As a result, they have a higher chance of accidents (Mayhew et al., 2002; Foss et al., 2011).

2.1.3 Environmental and Weather Factors

Studies by Adeyemi et al. (2023) and Chen et al. (2021) show that rain, snow, and fog increase crash risks. These conditions make roads slippery and reduce visibility. When combined with heavy traffic, they make accidents even worse (Adeyemi et al., 2023). Weather-related crashes account approximately 16% of all fatal crashes across United States whereas on average over 3,400 fatalities annually related to adverse weather conditions such as rain, snow, fog, and ice (Federal Highway Administration, 2025; Goldberg & Loren, 2024). 46% of weather-related crashes largely because of wet pavement conditions due to rain which consequently reduce traction and increase stopping distances (Federal Highway Administration, 2025; Goldberg & Loren, 2024). Winter weather increases on average accident rates by 34% and causes around 1,300 fatalities each year (Tobin et al., 2022; Goldberg & Loren, 2024). Visibility impairments from fog and heavy precipitation further increases crash risk (Tobin et al., 2022).

2.1.4 Advanced Analytical Techniques

Spatial autocorrelation techniques help analyze accident patterns. Two common methods are global Moran's I (Anselin, 1988) and Getis-Ord (Getis and Ord, 2010). Researchers suggest using spatial estimators to study crash injury risks (Jonathan et al., 2016; Lord et al., 2018). Tobler’s first law of geography states that all things are related, but nearby things are more related than distant things (Sui, 2004). Because of this, local Moran’s I (Anselin, 1995) and Getis-Ord GI* (Getis and Ord, 2010; Ord and Getis, 1995) are useful for finding accident

clusters (Waller and Gotway, 2004). Spatial autocorrelation generally weakens as the distance between locations increases (Epperson, 2005). Spatial models help understand accident patterns. They show that older drivers, especially those over 75, are more likely to be in fatal crashes at intersections. This remains true even when considering factors like time of day and weather (Preusser et al., 1997; Li et al., 2021). k-modes clustering and multiple correspondence analysis (MCA) has been used to uncover hidden patterns and relationships in traffic accident data, particularly when categorical variables are prevalent in crash datasets. It is used for categorizing traffic accidents factors into homogeneous groups based on factors such as weather, road geometry, time, and accident type (Prastya, Zulfadhilah, & Nurhaeni, 2021; Tesma, 2020)

2.2 Concluding remarks

This chapter gives the highlighting features of the previous studies related to this study. These studies showed what types of research had already been done and what to do in the future study.

Chapter 3
Methodology for the Research

3.1 Introduction

The step by step methodology, which combines data collection and processing, data transformation and cleaning, spatial and non-spatial analysis, for this research is given below:

3.2 Data Collection and Preprocessing

The FARS dataset, which includes detailed information on fatal crashes in all 50 states and the District of Columbia, was obtained from the NHTSA website. Crash sites in American Samoa, the Northern Mariana Islands, the U.S. Virgin Islands, Guam, and Puerto Rico were left out of the data because they are not part of the FARS national files for the 50 states and the District of Columbia. This set of data comes from police accident reports, death certificates, vehicle registration files, coroner reports, and records kept by emergency medical services. The dataset is open to the public and follows the Privacy Act, so it doesn't contain any information that could be used to find out who someone is. The data from 2015 to 2022 was first combined by year and then combined from all of those years. The combined data contain 308,704 people who died in 284,193 fatal crashes; for every year the number of fatalities matches the crash-level total (FATALS). The person, vehicle, accident and race files were merged in Python at the person level: each person was linked to the vehicle they were in (or, for pedestrians and cyclists, the vehicle that struck them), to the crash record, and to their own race record (keys ST_CASE, VEH_NO and PER_NO).

3.2.1 Data source and files

FARS is a census of fatal motor-vehicle crashes on public trafficways in the United States: a crash is included when at least one person died within 30 days of it. The national CSV files for

each year from 2015 to 2022 were downloaded from the NHTSA website. Each year contains about 30 linked files, of which four were used: the accident file (one row per crash), the vehicle file (one row per vehicle), the person file (one row per person involved, including survivors) and, from 2019, the race file (one row for each race listed for each person who died). The files are linked by three keys: ST_CASE identifies a crash within a year, VEH_NO a vehicle within the crash and PER_NO a person within the vehicle; pedestrians and other people outside a vehicle have VEH_NO = 0. Definitions and codes were taken from the FARS Analytical User's Manual (National Highway Traffic Safety Administration, 2024).

3.2.2 Building the person-level dataset

The unit of analysis is the person who died. The dataset was built for each year in five steps (Figure 3.1).

Step 1 – Selecting fatalities. From the person file, only people whose injury severity is coded as fatal (INJ_SEV = 4, "Fatal Injury (K)") were kept. Selecting by the numeric code rather than by the label text avoids errors when label wording changes between years.

Step 2 – Linking each person to a vehicle. FARS records the posted speed limit, number of lanes, trafficway description, alignment, grade, pavement type and surface condition for each vehicle, describing the road that vehicle was on just before the crash. A person who died inside a vehicle was linked to that vehicle (ST_CASE and VEH_NO). A pedestrian, cyclist or other person outside a vehicle was linked to the vehicle that struck them, given by the variable STR_VEH; when that vehicle is not known, the roadway variables were left unknown. Linking every person to the first vehicle of the crash instead would give the wrong road information whenever the vehicles in a crash were on different roads, for example at intersections.

Step 3 – Adding crash information. Variables that describe the whole crash were added from the accident file through ST_CASE: state, month, day of the week, functional system of the

road, rural or urban location, type of intersection, relation to the trafficway, light condition, weather, the number of people killed in the crash (FATALS) and the crash coordinates.

Step 4 – Adding race. Race is recorded only for people who died, from the death certificate. From 2019 it is stored in the separate race file, which can hold several races for one person; the first listed race (ORDER = 1) was used, as the FARS manual recommends for comparison with earlier years, and joined on ST_CASE, VEH_NO and PER_NO so that each person receives their own race. For 2015 to 2018 race was taken from the person file. Iowa's race data are redacted from 2020 and are therefore unknown.

Step 5 – Combining the years and validating. The eight yearly files were stacked into one file. Every merge was checked to be one-to-one or many-to-one, so that no person could be duplicated or lost, and for every year the number of fatalities in the person-level data was compared with the sum of FATALS in the accident file. All eight years match exactly (Table 3.2): the final dataset contains 308,704 people who died in 284,193 fatal crashes, and 20,015 crashes (7.0%) killed more than one person.

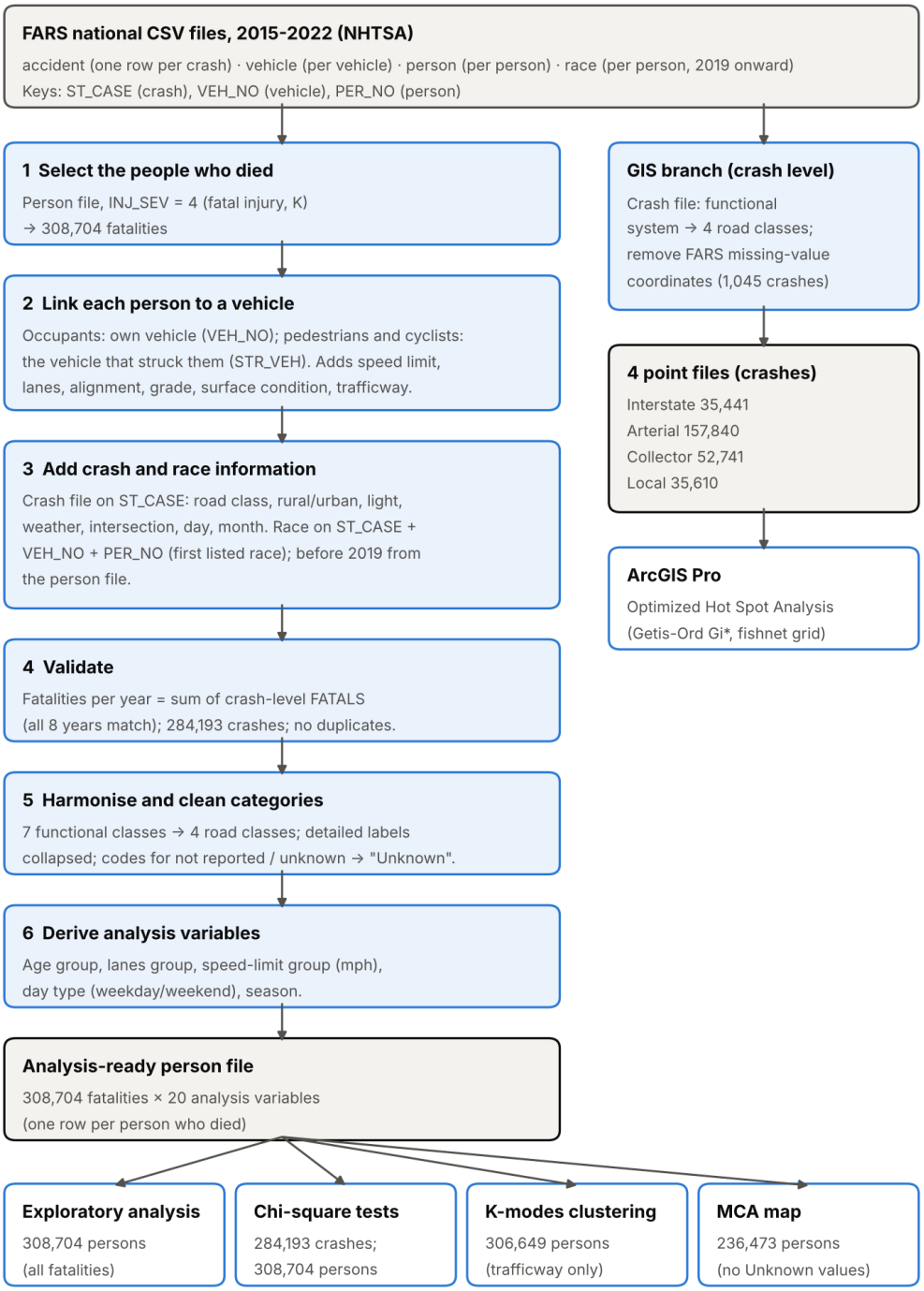


Figure 3.1 Workflow from the FARS Files to the Analysis-ready Datasets

Table 3.1 lists the variables used in the study and their categories after the cleaning described in Section 3.3.

Table 3.1 List of Variables and levels

Sl No	Variable	Variable Definition	Levels
1	**STATENAME**	State in which the accident occurred	Alabama, Alaska, Arizona, Arkansas, California, Colorado, Connecticut, Delaware, District of Columbia, Florida, Georgia, Hawaii, Idaho, Illinois, Indiana, Iowa, Kansas, Kentucky, Louisiana, Maine, Maryland, Massachusetts, Michigan, Minnesota, Mississippi, Missouri, Montana, Nebraska, Nevada, New Hampshire, New Jersey, New Mexico, New York, North Carolina, North Dakota, Ohio, Oklahoma, Oregon, Pennsylvania, Rhode Island, South Carolina, South Dakota, Tennessee, Texas, Utah, Vermont, Virginia, Washington, West Virginia, Wisconsin, Wyoming
2	**MONTHNAME**	Month in which the accident occurred	January, February, March, April, May, June, July, August, September, October, November, December
3	**DAY_WEEKNAME**	Day of the week of the accident	Thursday, Sunday, Wednesday, Saturday, Tuesday, Monday, Friday
4	**FUNC_SYSNAME**	Type of road (arterial, interstate, etc.)	Arterial, Collector, Interstate, Local, Unknown
5	**RACENAME**	Race of the person involved	White, Black, Asian, American Indian/Alaska Native, Other, Unknown
6	**RUR_URBNAME**	Rural or Urban area	Urban, Rural, Unknown
7	**SEXNAME**	Gender of the person involved	Male, Female, Other, Unknown
8	**PER_TYPNAME**	Type of person involved in the accident (e.g., driver, passenger)	Motor Vehicle Driver, Motor Vehicle Passenger, Pedestrian, Cyclist, Other, Unknown
9	**SEAT_POSNAME**	Position of the person in the vehicle	Front Seat, Not an occupant, Second Seat, Third/Fourth Seat, Other, Unknown
10	**VTRAFWAYNAME**	Type of trafficway	Two-Way Not Divided, Two-Way Divided, Ramp, One-Way Trafficway, Other, Unknown
11	**VALIGNNAME**	Type of alignment (e.g., curve, straight)	Straight, Curve - Left, Curve - Right, Curve - Unknown Direction, Other, Unknown

Sl No	Variable	Variable Definition	Levels
12	**VPROFILENAME**	Type of profile (e.g., level, uphill)	Level, Grade, Unknown Slope, Downhill, Uphill, Hillcrest, Sag, Other, Unknown
13	**VPAVETYPNAME**	Type of pavement	Blacktop, Bituminous, or Asphalt, Concrete, Other, Unknown
14	**VSURCONDNAME**	Surface condition (e.g., dry, wet, ice/frost)	Dry, Wet, Ice/Frost, Snow/Slush, Other, Unknown
15	**TYP_INTNAME**	Type of intersection	Not an Intersection, Intersection, Other, Unknown
16	**REL_ROADNAME**	Relative position to the road	On Roadway, On Roadside, On Median, On Shoulder, Other, Unknown
17	**LGT_CONDNAME**	Light condition at the time of accident	Daylight, Dark - Not Lighted, Dark - Lighted, Dawn/Dusk, Dark - Unknown Lighting, Other, Unknown
18	**WEATHERNAME**	Weather conditions at the time of accident	Clear, Cloudy, Rain, Snow, Fog, Other, Unknown
19	**AGE_GROUP**	Age group of the person who died (years)	0-15, 16-24, 25-44, 45-64, 65+, Unknown
20	**LANES_GROUP**	Number of lanes on the road	1-2 lanes, 3-4 lanes, 5-6 lanes, 7+ lanes, Non-trafficway/driveway, Unknown
21	**SPEED_LIMIT_GROUP**	Posted speed limit (mph)	5-25 mph, 30-40 mph, 45-55 mph, 60+ mph, No statutory limit, Unknown
22	**DAY_TYPE**	Day type (weekday or weekend)	Weekday, Weekend
23	**SEASON**	Season of the year	Winter, Spring, Summer, Fall

Table 3.2 Fatal Crashes and Fatalities by Year, 2015–2022

Year	Fatal crashes	Fatalities	Fatalities per crash	Fatalities = sum of FATALS
2015	32,538	35,484	1.09	Yes
2016	34,748	37,806	1.09	Yes
2017	34,560	37,473	1.08	Yes
2018	33,919	36,835	1.09	Yes
2019	33,487	36,355	1.09	Yes
2020	35,935	39,007	1.09	Yes
2021	39,785	43,230	1.09	Yes
2022	39,221	42,514	1.08	Yes
Total	**284,193**	**308,704**	**1.09**	

3.3 Data Cleaning and Transformation

The FARS dataset includes different types of category variables with multiple levels. The data encompasses fatal traffic collisions, indicating that at least one individual died in the accident. To enhance analysis and interpretation, variables' levels were changed by decreasing the number of levels for each variable. For example, the seven FHWA functional classes were grouped into four road classes: Interstate, Arterial (principal arterials and minor arterials), Collector (major and minor collectors) and Local. Moreover, continuous integer variables, such as age, number of lanes and posted speed limit, were grouped into ranges: age 0–15, 16–24, 25–44, 45–64 and 65 and over; lanes 1–2, 3–4, 5–6 and 7 or more; speed limit 5–25, 30–40, 45–55 and 60 mph or more. These changes were done to enhance the manageability and interpretability of the data for clustering and statistical analysis.

FARS uses numeric codes for values that are not reported or unknown: 998 and 999 for age, 8 and 9 for the number of lanes, 98 and 99 for the speed limit, and 77.7777, 88.8888 and 99.9999 for latitude (777.7777 and so on for longitude) (National Highway Traffic Safety Administration, 2024). These codes were recoded as “Unknown” before any grouping. In the first version of this analysis they had been grouped as real values: 1,040 people with unknown age were counted as “Over 67”, 3,297 people with unknown lane counts made up 54% of the “7+ lanes” group, and 11,420 people with unknown speed limits were counted in the “60–100” group. Code 0 for lanes and speed limit (non-trafficway or no statutory limit) is kept as its own category. Labels such as “Not Reported” and “Reported as Unknown” are kept as “Unknown” rather than merged into “Other”.

3.3.1 Harmonising categories across years

FARS labels change between years (for example “Unknown” became “Reported as Unknown” in 2018, and new categories were added to several variables), so categories were mapped with

keyword rules that work for every year. Detailed categories were then collapsed to the levels in Table 3.1. For example, the seven functional classes were grouped into the four road classes; the Chinese, Japanese, Filipino, Korean, Vietnamese, Asian Indian and other Asian race codes into Asian; the two kinds of divided trafficway (with an unprotected median or with a median barrier) into Two-Way Divided; and snow and slush on the road into Snow/Slush. Values recorded as not reported or unknown were kept as a separate category, "Unknown", and never merged with "Other", so that their share stays visible.

3.3.2 Derived variables

Five variables were derived. Age was grouped into 0–15, 16–24, 25–44, 45–64 and 65 years and over. The number of lanes was grouped into 1–2, 3–4, 5–6 and 7 or more, with a separate category for non-trafficway areas and driveways (code 0). The posted speed limit, which FARS records in 5-mph steps, was grouped into 5–25, 30–40, 45–55 and 60 mph or more, with a separate category for no statutory limit (code 0). The day of the week was grouped into weekday (Monday to Friday) and weekend (Saturday and Sunday), and the month into seasons (winter: December to February; spring: March to May; summer: June to August; fall: September to November).

3.3.3 Data quality

After cleaning, the share of unknown values is below 2% for most variables but higher for pavement type (25.3%), grade (8.6%), race (6.1%), weather (5.4%) and speed limit (3.9%). Because a quarter of its values are unknown, pavement type was not used in the cluster analysis. Seat position was also left out of the clustering, because for pedestrians and cyclists it only repeats the person type.

3.3.4 Crash locations for the spatial analysis

For the hot spot analysis, the accident file was used directly (one row per crash). Crashes were assigned to the four road classes; 1,516 crashes whose functional system was not in the state inventory, not reported or unknown were left out, as were 1,045 crashes whose coordinates carry FARS missing-value codes. The four point files contain 35,441 interstate, 157,840 arterial, 52,741 collector and 35,610 local crashes (281,632 in total), and were imported into ArcGIS Pro from their latitude and longitude.

3.3.5 Analysis-ready datasets

Table 3.3 summarises the datasets used in Chapter 4. Each analysis uses the unit that matches its question: person-level summaries count every person who died, while variables that describe the crash or the road are tested with one row per crash, so that a crash in which several people died is not counted several times.

Table 3.3 Analysis-ready Datasets

Dataset	Unit	Rows	Used for
Person-level file	Person who died	308,704	Exploratory analysis (4.2); person-level chi-square tests
Crash-level file	Fatal crash	284,193	Chi-square tests of road and environment variables
Clustering file	Person who died on a trafficway	306,649	K-modes clustering (18 variables)
MCA file	Person with no unknown value	236,473	MCA map of the clusters
Crash point files	Fatal crash with valid coordinates	281,632	Optimized Hot Spot Analysis (4 road classes)

3.4 Chi-square Test of Association

The chi-square test of independence was applied here to know the relationship between the target variable which is road hierarchy and various categorical features. The null hypothesis for this examination is : There is no significant relationship between the target variable and the feature variables. The alternative hypothesis is that there is significant relationship between

target variable and various categorical features. Here is the formula that we measure this statistics:

$$\chi^2 = \sum_i \frac{(O_i - E_i)^2}{E_i} \qquad \text{..... (3.1)}$$

Where O_i is the observed frequency in the i-th category and E_i is the expected frequency in the i-th category. This is calculated under the assumption of independence.

The chi-square statistic is then compared to the critical value from the chi-square distribution table. This will then determine the significance level. If the p-value is below a given threshold which is usually 0.05, the null hypothesis is rejected. This indicates a significant relationship between the variables (Agresti, 2013).

Road and environment variables describe the crash, so they were tested with one row per crash (about 282,000 crashes); person variables (person type, seat position, age, race and sex) were tested with one row per fatality. Unknown values were excluded pair by pair. Because the samples are very large, almost every test is significant, so the strength of each association was judged with the bias-corrected Cramér's V (Bergsma, 2013) rather than with the p-value.

3.5 Spatial Analysis: Hot Spot Analysis Based on Road Hierarchy

To explore the spatial distribution of fatal crashes, a hot spot analysis was performed using ArcGIS Pro. The crash locations were categorized based on the type of road which are Interstate, Arterial, Collector, Local Roads.

Optimized Hot Spot Analysis was performed using the *Getis-Ord Gi statistic**, This statistics identifies statistically significant clusters of high and low crash rates. Here is the formula for the *Getis-Ord Gi statistic** :

..... (3.2)

$$G_i(d) = \frac{\sum_j w_{ij}(d) x_j}{\sum_j x_j}$$

Here , the x_j are the weighted values of the points in the study area. W_{ij} a binary, symmetric weights matrix with ones for all points j within distance d of point i and zeros otherwise. There are two variants of the local G statistic. The Gi* includes the value at i in the summation and is most often used for studies of clustering.

Essentially, the Gi statistic represents a ratio of the values within distance d of point i to the sum of all values minus the value at point i. The Gi* includes all values. The analysis was conducted on a Fishnet grid, a grid-based approach that divides the study area into smaller cells to aggregate the crash data. The analysis generates maps showing high-crash areas (hot spots) and low-crash areas (cold spots), which are then examined to understand the spatial patterns of fatal crashes across different road types. The tool chose the cell size separately for each road class (about 0.67 to 0.76 degrees per side). Because no bounding polygon was supplied, only cells containing at least one crash were analysed, so areas without any crash of that road class appear blank and are not tested. The False Discovery Rate correction for multiple testing was applied. The analysis used counts of crashes per cell without a measure of exposure, so hot spots show where fatal crashes are concentrated, not where the risk per vehicle-mile is highest.

3.6 Non-Spatial Analysis: K-Mode Clustering

K-mode clustering which is a non-spatial tool to group data into k clusters. Theses cluster are formed based on similarity in the feature space. In this study, the features used for clustering included road type, weather conditions, age, sex, and other roadway characteristics. K-modes (Huang, 1998) adapts k-means to categorical data: the distance between two records is the

number of variables on which they differ (simple matching dissimilarity), each cluster is represented by its mode (the most frequent category of each variable), and records are reassigned to the nearest mode until the total dissimilarity stops decreasing.

Here is the K-mode objective function:

$$P(W, Q) = \Sigma l=1..k \, \Sigma i=1..n \, w_{il} \, d(X_i, Q_l) \quad \text{with} \quad d(X, Y) = \Sigma j=1..m \, \delta(x_j, y_j), \quad \delta = 0 \text{ if the categories are equal, 1 otherwis} \quad \ldots\ldots\ldots\ldots\ldots\ldots \quad (3.3)$$

where X_i is the i-th record, Q_l is the mode of cluster l, w_il equals 1 if record i belongs to cluster l and 0 otherwise, and m is the number of variables.

The number of clusters k was chosen with three diagnostics computed for k = 1 to 8 on a random sample of 60,000 records with five random starts for each k: the elbow of the K-modes cost, the mean silhouette width computed with the Hamming (simple matching) distance (Rousseeuw, 1987), and the stability of the solution across random starts, measured with the adjusted Rand index (Hubert & Arabie, 1985). The final model was fitted on all records with a fixed random seed. Eighteen variables were used; pavement type (25% unknown) and seat position (which repeats person type for pedestrians and cyclists) were left out, and 2,055 deaths on non-trafficway areas or driveways were excluded. Clusters were described by the categories that are over-represented in each cluster (share in the cluster compared with the share among all fatalities), because the most frequent category of many variables, such as male, clear weather or dry surface, is the same in every cluster.

3.7 Results and Interpretation

The results which is got from the spatial and non-spatial analysis, this will give a comprehensive understanding of the factors that are influencing fatal crashes in different states of US. The spatial analysis identified areas with high and low concentration of fatal crashes

and the non-spatial clustering analysis form unique group which will show different common characteristics that are related to road types, weather conditions, and demographic factors. These findings will shape up the guide policy recommendations for improving road safety.

3.8 Analytical tools and software

ArcGIS Pro was utilized for spatial data processing like geocoding crash location and optimized hot spot analysis. For statistical and clustering analysis, Python 3 was used with pandas for data processing, SciPy for the chi-square tests, scikit-learn for the silhouette and adjusted Rand index, kmodes for K-modes clustering, prince for multiple correspondence analysis and matplotlib for figures. All code is available at https://github.com/<username>/fars-fatal-crash-clustering and is described in the Appendix.

3.9 Concluding remarks

This chapter discusses about the in detailed methodology of this study to have a clear idea about the data collection and cleaning process and how the results are generated of this study.

Chapter 4
Data Analysis and Results

4.1 Introduction

In this chapter, we will discuss about the exploratory data analysis, spatial and non-spatial cluster analysis and results of these analysis

4.2 Exploratory data analysis

4.2.1 Geographical Context

The USA map here shows the boundaries of all 50 states and District of Columbia (Washington D.C) . The main purpose of this map is to provide geographical context for understanding where accident happens in the following charts. It is helpful for linking up state-level crash data and trends with specific locations across the country.

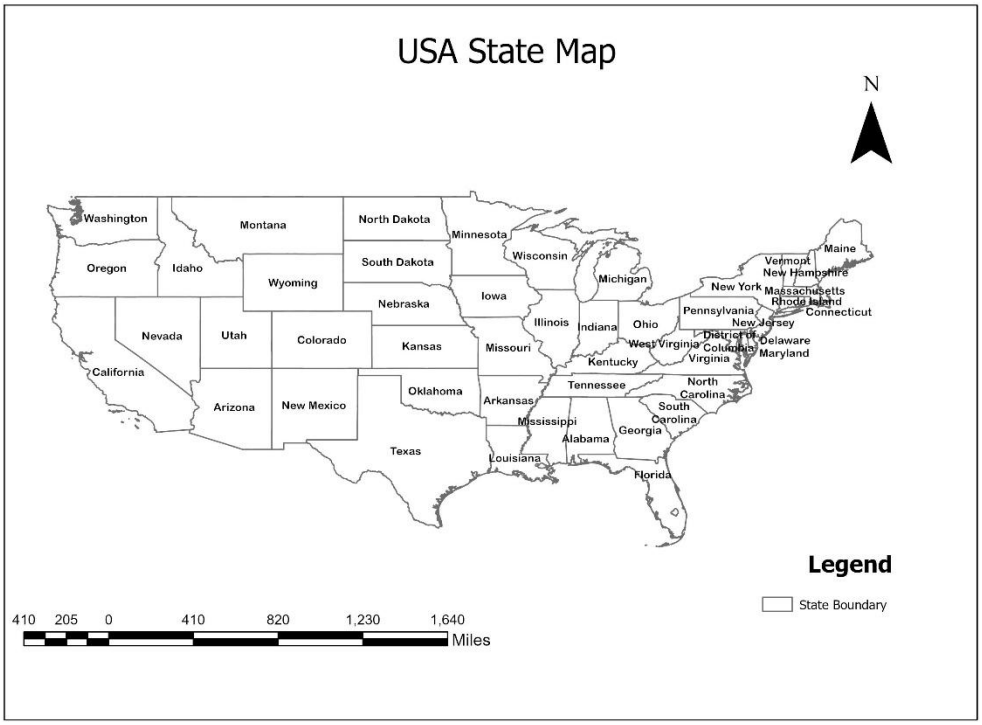


Figure 4.1 USA State Map Source: (U.S. Census Bureau, 2025)

4.2.2 State-wise Yearly Percentage Change in Fatalities (Year-over-Year)

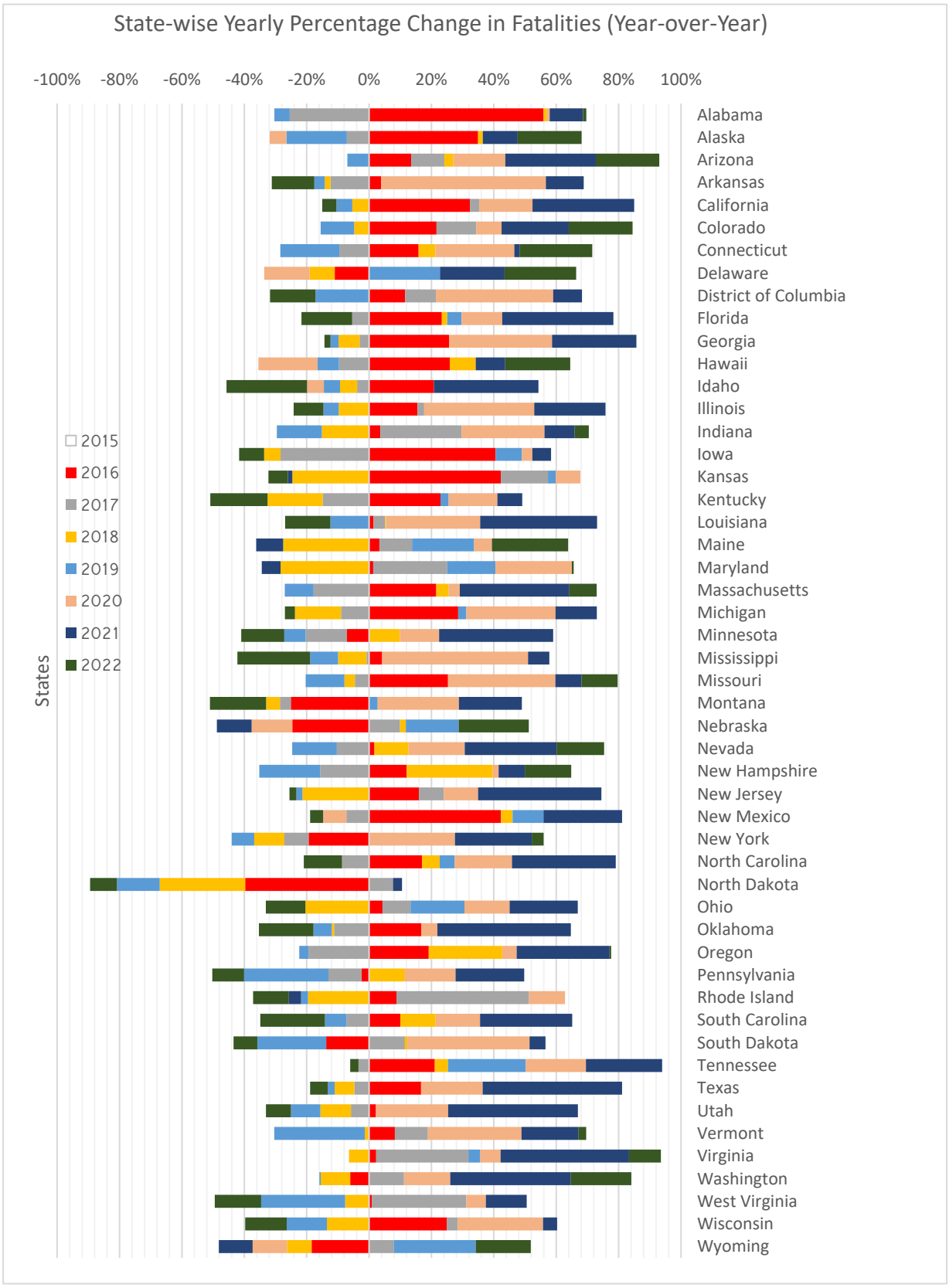


Figure 4.2 State-wise Yearly Percentage Change in Fatalities (Year-over-Year)

The State-wise Yearly Percentage Change in Fatalities from 2015 to 2022 figure provides a comprehensive analysis of the variation in fatalities across all the mentioned states of U.S. Each state is represented by a horizontal bar. The length of the bar displays the percentage change in fatalities relative to the previous year. Different colors on the bars represent each year such as red for 2016, blue for 2022).

The chart highlights notable increase or decrease of fatalities over the years. For example, Alabama and Iowa experienced some of the highest percentage changes in 2016 which shows a higher volatility in traffic fatalities year-to-year. The year 2020 and 2021 stands out across multiple states. There is a spike in fatalities. The COVID-19 pandemic can be responsible as it led to reduced traffic volumes on the roads but also an increase in risky driving behaviors. It contributed to increased accidents and fatalities in certain areas. In many states, 2022 show a gradual reduction of fatalities compared to 2021. States like California and Florida exhibit less volatility year-over-year changes which represents more stable trends. However, certain states like Mississippi and Louisiana show more volatile shifts which indicates that these states may have experienced road safety challenges that fluctuate significantly over time.

Some states show decreases in most years, such as North Dakota. On the other hand, certain states, including Arizona and Tennessee, show increases in most years. States with larger negative bars , for example Vermont in 2019, indicate a decrease in fatalities in certain years. States with larger positive bars suggest worsening trends in road safety.

4.2.3 Seasonal Fatal crashes Trends (2015-2022)

The Seasonal Fatal Crash Trends (2015-2022) chart illustrates how the average number of fatalities per day differs across different seasons (Spring, Summer, Fall, and Winter) from 2015 to 2022.

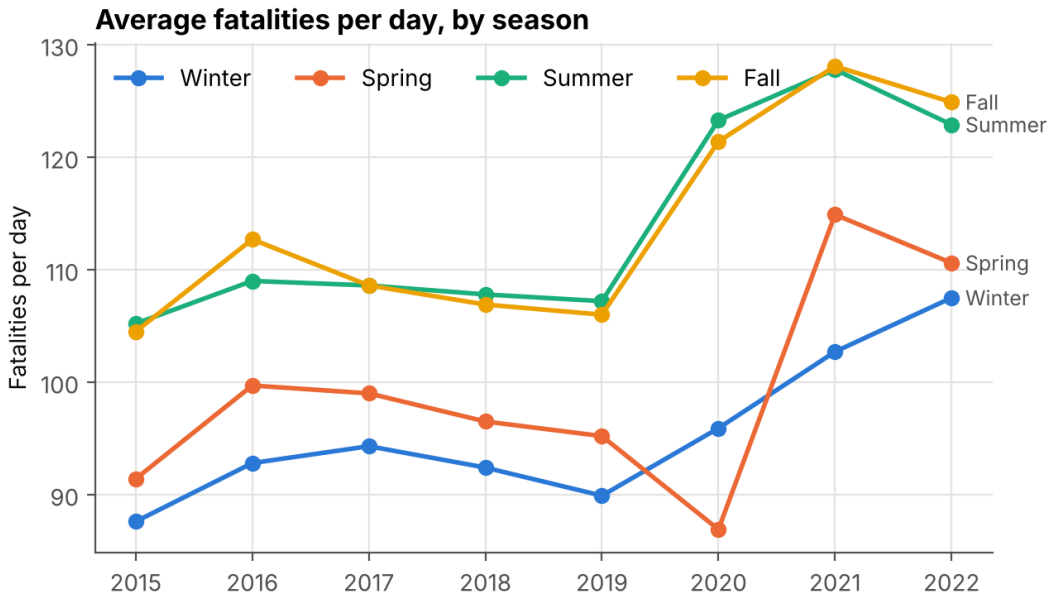


Figure 4.3 Average Fatalities per Day by Season (2015-2022)

This line graph helps to find seasonal patterns. It represents the highest and lowest number of fatalities that occurred over the specified period. If the month falls within "March", "April", or "May", it is classified as "Spring". "Summer" is assigned to the months of "June", "July", and "August". For "September", "October", or "November", the month is considered "Fall". The months of "December", "January", and "February" are grouped under "Winter".

Summer and fall have the most fatalities per day in every year (between 104 and 128 per day), and spring and winter the fewest. Fall is slightly higher than summer in 2016, 2021 and 2022.

Fatalities per day rose sharply in 2020 and reached their highest level in 2021 in every season. The exception is spring 2020, which fell to 87 per day, the lowest value of the period, during the first months of the COVID-19 pandemic, before rising again in summer 2020. Winter rose steadily from 90 per day in 2019 to 108 in 2022.

These figures are counts. Seasonal differences in the amount of travel, which FARS does not record, are a likely part of the explanation, so explanations such as holiday travel or school terms remain possible reasons rather than findings of this study.

4.2.4 Fatal Crash Count Based on Weekday and Weekend

This bar chart compares the number of fatalities on weekdays and at weekends. It shows the total fatalities for each day of the week. Saturday has the most fatalities (17.5%), followed by Sunday (15.7%) and Friday (15.5%). People tend to enjoy more leisure travel, social gatherings, and long-distance trips in weekends Weekdays like Thursday and Friday also show high crash rates, indicating that commuters and after-work traffic contribute to the accidents during these days. Mondays and Tuesdays have fewer accidents. It may be because fewer vehicles on the road compared to weekend traffic.. Because a week has five weekdays and two weekend days, the comparison should be made per day: an average weekend day has about 24% more fatalities than an average weekday.

4.2.5 Fatal Crash Count Based on Road Hierarchy

This chart compares the number of fatalities by the type of road hierarchy: Arterial Roads, Collector Roads, Interstate Roads, Local Roads, and Unknown (0.52%). Arterial Roads account for the largest percentage (55.86%) of all fatalities. Because these roads are high-traffic thoroughfares connecting urban and rural areas. Collector Roads contribute 18.45% of crashes. It may be due to their role in connecting smaller roads to major highways. Local Roads contribute 12.24%. Interstate Roads account for 12.93% as these roads have higher speeds and long-distance traffic.

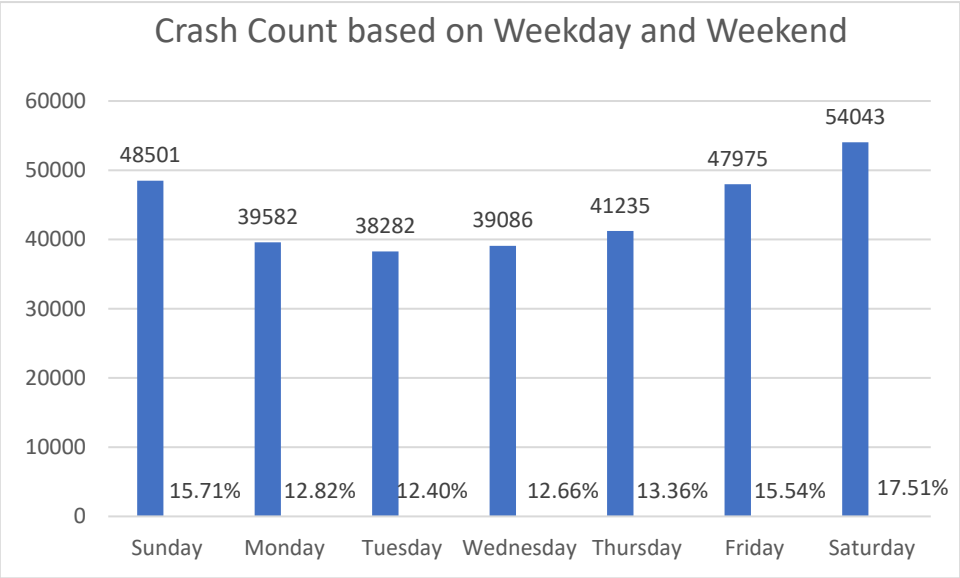


Figure 4.4 Crash Count Based on Weekday and Weekend

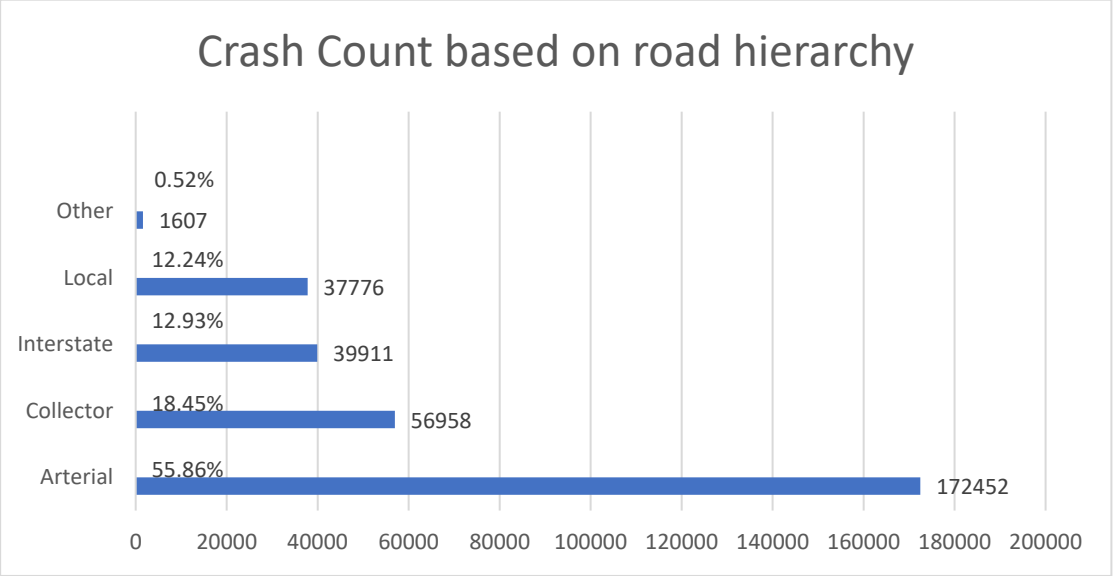


Figure 4.5 Crash Count Based on Road Hierarchy

4.2.6 Top Ten and Bottom Ten Fatal Crash rates per 100,000 Population

This table presents the average fatality rates per 100,000 population across states in the U.S. It ranked from the highest to the lowest accident rates. The Top Ten States with the highest accident rates are: Mississippi (23.5), Wyoming (21.9), South Carolina (20.5), and others. These states show consistently high accident rates. It could be related to various types of factors like poor infrastructure, higher speed limits, or environmental factors.

The Bottom Ten States with the lowest accident rates (e.g., Utah with 8.7) may have better traffic management and strict driving laws. The data suggests a distinctive difference in safety across states. The higher rates are concentrated in certain regions like the South and Midwest.

Table 4.1 Top Ten and Bottom Ten Fatal Crash rates per 100,000 Population

Top ten fatal crash rates per 100,000 population per year				**Bottom ten fatal crash rates per 100,000 population per year**		
Serial	**State Name**	**Average Accident rate**		**Serial**	**State Name**	**Average Accident rate**
1	Mississippi	23.5		1	Utah	8.7
2	Wyoming	21.9		2	Connecticut	8.2
3	South Carolina	20.5		3	Washington	7.6
4	Arkansas	19.3		4	Hawaii	7.2
5	New Mexico	19.2		5	Minnesota	7.1
6	Alabama	19.1		6	New Jersey	6.6
7	Montana	18.8		7	Rhode Island	5.4
8	Louisiana	17.4		8	Massachusetts	5.3

Top ten fatal crash rates per 100,000 population per year				Bottom ten fatal crash rates per 100,000 population per year		
Serial	**State Name**	**Average Accident rate**		**Serial**	**State Name**	**Average Accident rate**
9	Kentucky	17.1		9	New York	5.2
10	Oklahoma	17.1		10	District of Columbia	4.4

4.2.7 Weather, Light, and Surface Condition-Based Fatal Accident Distribution

This table shows the relationship between weather conditions, light conditions, and surface conditions at the time of the fatal crashes. The majority of fatal accidents occur in clear weather (70.66%), particularly under daylight conditions (33.24% of all fatalities occurred in clear weather and daylight). Because most driving takes place in clear weather, this share mainly reflects exposure and does not show that weather is unimportant. Rain accounts for 7.2% and snow for 1.2% of fatalities; wet surfaces for 11.7% and ice, frost or snow on the road for 1.8%. Dawn and dusk, which were previously part of "Other", account for 4.3%.

Table 4.2 Weather, Light and Surface Condition-Based Fatal Accident Distribution

Light / surface	Clear	Cloudy	Rain	Snow	Fog	Other/Unknown	Grand Total
Daylight	**33.24%**	**6.94%**	**3.00%**	**0.58%**	**0.24%**	**2.64%**	**46.63%**
Dry	31.79%	5.50%	0.01%	0.00%	0.14%	2.30%	39.74%
Wet	0.38%	1.11%	2.89%	0.11%	0.07%	0.10%	4.66%
Ice/Frost/Snow	0.18%	0.14%	0.05%	0.45%	0.01%	0.05%	0.87%
Other/Unknown	0.89%	0.19%	0.05%	0.02%	0.01%	0.19%	1.35%
Dark - Not Lighted	**18.97%**	**3.98%**	**2.23%**	**0.37%**	**0.59%**	**1.39%**	**27.52%**
Dry	18.16%	2.98%	0.01%	0.01%	0.34%	1.16%	22.65%
Wet	0.39%	0.85%	2.16%	0.07%	0.22%	0.10%	3.79%
Ice/Frost/Snow	0.13%	0.09%	0.04%	0.28%	0.02%	0.04%	0.60%
Other/Unknown	0.28%	0.06%	0.02%	0.01%	0.01%	0.09%	0.49%
Dark - Lighted	**14.83%**	**2.45%**	**1.61%**	**0.14%**	**0.17%**	**0.88%**	**20.08%**
Dry	14.25%	1.86%	0.00%	0.00%	0.09%	0.76%	16.97%
Wet	0.29%	0.52%	1.58%	0.05%	0.07%	0.05%	2.56%
Ice/Frost/Snow	0.05%	0.03%	0.01%	0.08%	0.00%	0.02%	0.20%
Other/Unknown	0.23%	0.04%	0.02%	0.01%	0.00%	0.05%	0.35%

Light / surface	Clear	Cloudy	Rain	Snow	Fog	Other/Unknown	Grand Total
Dawn/Dusk	**2.88%**	**0.67%**	**0.31%**	**0.06%**	**0.11%**	**0.21%**	**4.26%**
Dry	2.74%	0.50%	0.00%	0.00%	0.07%	0.19%	3.50%
Wet	0.05%	0.15%	0.30%	0.01%	0.04%	0.01%	0.56%
Ice/Frost/Snow	0.02%	0.02%	0.01%	0.05%	0.00%	0.01%	0.10%
Other/Unknown	0.06%	0.01%	0.00%	0.00%	0.00%	0.01%	0.09%
Other/Unknown	**0.74%**	**0.09%**	**0.09%**	**0.01%**	**0.02%**	**0.56%**	**1.51%**
Dry	0.69%	0.07%	0.00%	0.00%	0.01%	0.09%	0.86%
Wet	0.02%	0.02%	0.08%	0.00%	0.00%	0.01%	0.14%
Ice/Frost/Snow	0.00%	0.00%	0.00%	0.01%	0.00%	0.01%	0.02%
Other/Unknown	0.03%	0.00%	0.00%	0.00%	0.00%	0.45%	0.49%
Grand Total	**70.66%**	**14.13%**	**7.23%**	**1.17%**	**1.12%**	**5.68%**	**100.00%**

4.2.8 Age, Race, and Sex-Based Fatal Accident Distribution

This table shows the distribution of fatal accidents based on age, sex, and race. It provides insights into the demographic characteristics of people involved in accidents. The largest age group is 25–44 (33.1% of fatalities), followed by 45–64 (28.2%), 65 and over (18.2%) and 16–24 (16.8%). Per year of age, the 16–24 band has the largest share (1.9% of fatalities per year of age, compared with 1.7% for 25–44 and 1.4% for 45–64). Men account for 71.3% of fatalities and women for 28.5%; men aged 25–44 alone account for 24.6%.

By race, 72.1% of those who died were recorded as White, 16.0% as Black, 2.2% as Asian, 1.7% as American Indian or Alaska Native and 2.0% as other races; race is unknown for 6.1%, including all Iowa deaths from 2020, when Iowa's race data were redacted. These shares broadly follow the composition of the population, so without population denominators they do not show that any group is at higher or lower risk.

Table 4.3 Age, Race, and Sex-Based Fatal Accident Distribution

Age group / race	Female	Male	Other/Unknown	Grand Total
0-15	**1.49%**	**1.94%**	**0.01%**	**3.44%**
White	0.93%	1.22%	0.00%	2.16%
Black	0.34%	0.45%	0.00%	0.79%
Asian	0.03%	0.04%	0.00%	0.08%
Other (incl. AIAN)	0.10%	0.11%	0.00%	0.21%
Unknown	0.09%	0.11%	0.00%	0.20%
16-24	**4.83%**	**11.91%**	**0.01%**	**16.75%**
White	3.35%	8.36%	0.01%	11.72%
Black	0.88%	2.02%	0.00%	2.90%

Age group / race	Female	Male	Other/Unknown	Grand Total
Asian	0.10%	0.26%	0.00%	0.36%
Other (incl. AIAN)	0.25%	0.58%	0.00%	0.83%
Unknown	0.26%	0.68%	0.00%	0.94%
25-44	**8.51%**	**24.56%**	**0.03%**	**33.10%**
White	5.86%	16.74%	0.02%	22.61%
Black	1.58%	4.78%	0.01%	6.37%
Asian	0.18%	0.49%	0.00%	0.67%
Other (incl. AIAN)	0.43%	1.09%	0.00%	1.52%
Unknown	0.47%	1.46%	0.00%	1.93%
45-64	**7.24%**	**20.87%**	**0.03%**	**28.15%**
White	5.39%	15.40%	0.02%	20.81%
Black	1.03%	3.26%	0.01%	4.30%
Asian	0.21%	0.35%	0.00%	0.56%
Other (incl. AIAN)	0.21%	0.58%	0.00%	0.79%
Unknown	0.41%	1.28%	0.00%	1.69%
65+	**6.35%**	**11.86%**	**0.02%**	**18.22%**
White	5.09%	9.49%	0.01%	14.59%
Black	0.50%	1.08%	0.00%	1.58%
Asian	0.26%	0.28%	0.00%	0.55%
Other (incl. AIAN)	0.10%	0.19%	0.00%	0.29%
Unknown	0.41%	0.81%	0.00%	1.22%
Unknown	**0.07%**	**0.19%**	**0.08%**	**0.34%**
White	0.03%	0.10%	0.04%	0.17%
Black	0.02%	0.04%	0.02%	0.08%
Asian	0.00%	0.00%	0.00%	0.01%
Other (incl. AIAN)	0.00%	0.01%	0.00%	0.01%
Unknown	0.01%	0.04%	0.02%	0.07%
Grand Total	**28.49%**	**71.34%**	**0.17%**	**100.00%**

4.3 Spatial and Non Spatial cluster Analysis

4.3.1 Testing the Independent Association of road hierarchy

The Chi-Square Statistic tests, which is described earlier, compares the observed frequency distribution of a variable to the expected distribution. A large chi-square value gives evidence that the variables are not independent, but its size also grows with the number of observations, so it does not measure how strong the association is.

Cramér's V is a related tool of Chi square statistics. This study reports the bias-corrected version of Cramér's V (Bergsma, 2013). It usually measure of association for categorical variables. It quantifies the strength of association between two categorical variables. This can have range from 0 to 1. Here ‘0’ means no association and ‘1’ means a very strong association. Here’s a general rule of thumb for interpreting Cramér's V: 0 to 0.1 defines as Very weak

association, 0.1 to 0.3 defines as Weak association, 0.3 to 0.5 defines as Moderate association, 0.5 to 1.0 is defined as Strong association.

Results

The results of the Chi-Square tests and the corresponding Cramér's V values for all variables are summarized in the table below. Every test was significant ($p < 0.001$), which is expected with more than 280,000 crashes; the strength of each association is therefore judged by Cramér's V.

Table 4.4 Chi Square test results

Variable	Unit	n	Chi-square	df	p-value	Cramér's V (bias-corrected)	Strength
Speed limit group	Crash	271,928	111,849	12	<0.001	0.370	Moderate
Trafficway type	Crash	280,959	97,700	12	<0.001	0.340	Moderate
Rural/urban	Crash	282,574	15,827	3	<0.001	0.237	Weak
Pavement type	Crash	211,604	22,604	6	<0.001	0.231	Weak
Number of lanes	Crash	279,515	34,272	12	<0.001	0.202	Weak
Relation to road	Crash	281,313	31,635	12	<0.001	0.194	Weak
Intersection type	Crash	281,951	14,780	6	<0.001	0.162	Weak
Alignment	Crash	276,259	7,961	12	<0.001	0.098	Negligible
Light condition	Crash	281,161	5,872	15	<0.001	0.083	Negligible
Person type	Fatality	306,563	4,732	12	<0.001	0.072	Negligible
Seat position	Fatality	304,549	4,565	12	<0.001	0.071	Negligible
Surface condition	Crash	278,296	2,365	12	<0.001	0.053	Negligible
Race	Fatality	288,532	2,439	12	<0.001	0.053	Negligible
Age group	Fatality	306,073	2,496	12	<0.001	0.052	Negligible
Grade/profile	Crash	258,370	2,041	18	<0.001	0.051	Negligible
Day type (weekday/weekend)	Crash	282,677	277	3	<0.001	0.031	Negligible
Weather	Crash	267,508	520	15	<0.001	0.025	Negligible
Sex	Fatality	306,584	289	6	<0.001	0.021	Negligible
Season	Crash	282,677	225	9	<0.001	0.016	Negligible

Road class is moderately associated with the posted speed limit ($V = 0.37$) and the trafficway type ($V = 0.34$), and weakly associated with rural or urban location (0.24), pavement type (0.23), number of lanes (0.20), relation to the roadway (0.19) and intersection type (0.16). The characteristics of the person who died (person type, seat position, age, race and sex), the weather, surface condition, season and day type are only negligibly associated with road class ($V < 0.10$). Road class therefore stands mainly for road design and speed, not for who is killed.

4.3.2 Spatial Clustering (Hot Spot analysis)

This study tries to explore the spatial distribution of crash locations using 'Hot-spot analysis', a distinctive tool to identify concentrated area of fatal accidents, based on the road hierarchies. There are mainly 4 types of roads in USA : Arterial roads, Collector Roads, Interstate Roads and Local Roads. The database for fatal crashes has been separated based on road types for 2015-2022. This tool can assess the areas that need prioritized intervention using different levels of confidence. This spatial analysis differentiates 'Hot Spot' and 'Cold Spot' based on the aggregate occurrence of fatal crashes based on fishnet grid. Both 'Hot Spot' and 'Cold Spot' have three stages of confidence level: 99%, 95% and 90% to make it more distinctive. Because the analysis uses counts of crashes per cell, hot spots show where fatal crashes are concentrated, which largely follows population and traffic volume. They do not by themselves show where the risk per vehicle-mile is higher, and cold spots do not mean that no intervention is needed.

1. Analysis for Arterial Road

Arterial road is defined as the primary road that is designed for high-speed, long distance travel and major road connecting different regions within a state. After conducting Hot Spot analysis on arterial roads, it was found that northeast and southeast regions were showing prominent hot spots with 99% confidence level which included Pennsylvania, New York, New Jersey, Florida, Georgia, and Alabama. Some parts of California and Texas also presented notable concentrations of crashes. The northern part of USA were showing cold spots (at the 90% and 95% confidence levels) including Montana, Idaho, Wyoming, North Dakota and South Dakota**.**

2.Analysis for Collector Road

Collector road is defined as connectors between local roads and arterial roads and also characterized by moderate speeds and traffic volumes. Collector roads show a broader distribution of hot spots across the U.S. Almost all of the eastern part of the U.S. specially in states like Tennessee, Indiana, Kentucky, North Carolina, South Carolina, Pennsylvania, , New Jersey, Florida, Georgia, Alabama, Louisiana, Virginia and parts of Texas and California showed hot spots with 99% confidence level. Th middle part of U.S. particularly states like Colorado, Kansas, New Mexico, Montana, Idaho, Wyoming, North Dakota and South Dakota showed cold spot with a 99% of confidence level.

3.Analysis for Interstate road

Interstate roads are mainly used for state to state travel with higher speed limits and more extensive infrastructure. For this type of road, the concentration of the fatal crashes in specific area is lower. California, Texas and Florida stand out as regions with significant crash concentration on interstate highways. Of the 990 cells analysed, 620 were not statistically significant, 188 were hot spots and 182 cold spots. The blank areas on the map are places with no interstate fatal crash: without a bounding polygon, the tool analyses only cells that contain at least one crash.

4. Analysis for Local Road

Local roads are smaller roads that serve local traffic, having lower speed limits and present higher accident risks as non-motorized vehicles are present on the road. Analysis on local roads is almost similar to collector roads which showed a broader distribution of hot spots across the U.S. Almost all of the eastern part of the U.S. specially in states like Tennessee, Indiana,

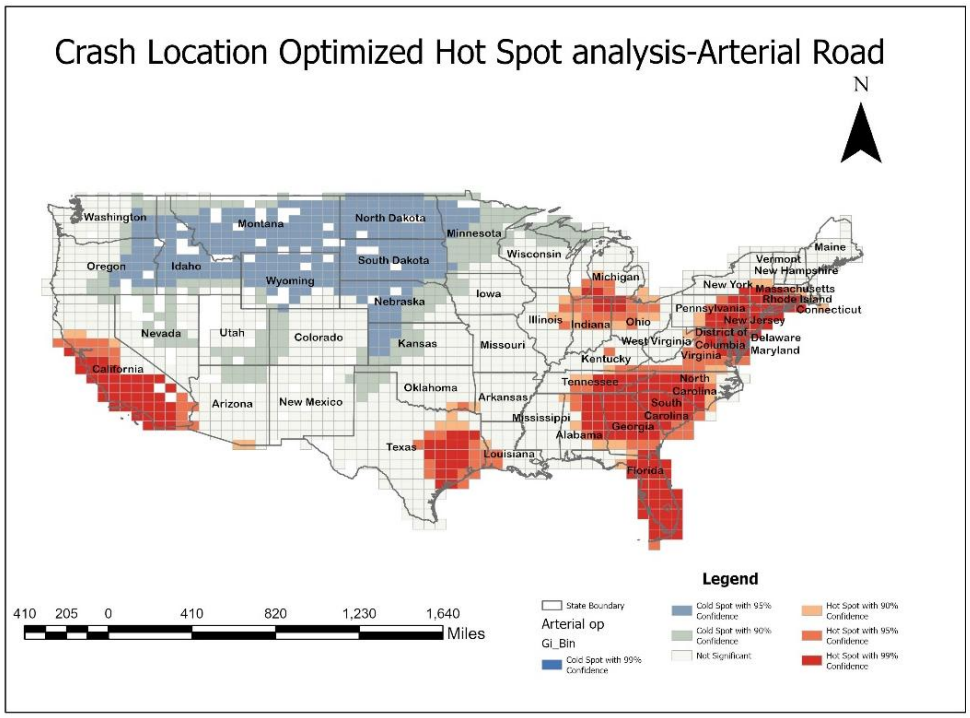


Figure 4.6 Crash location Hot Spot analysis based on Arterial Road

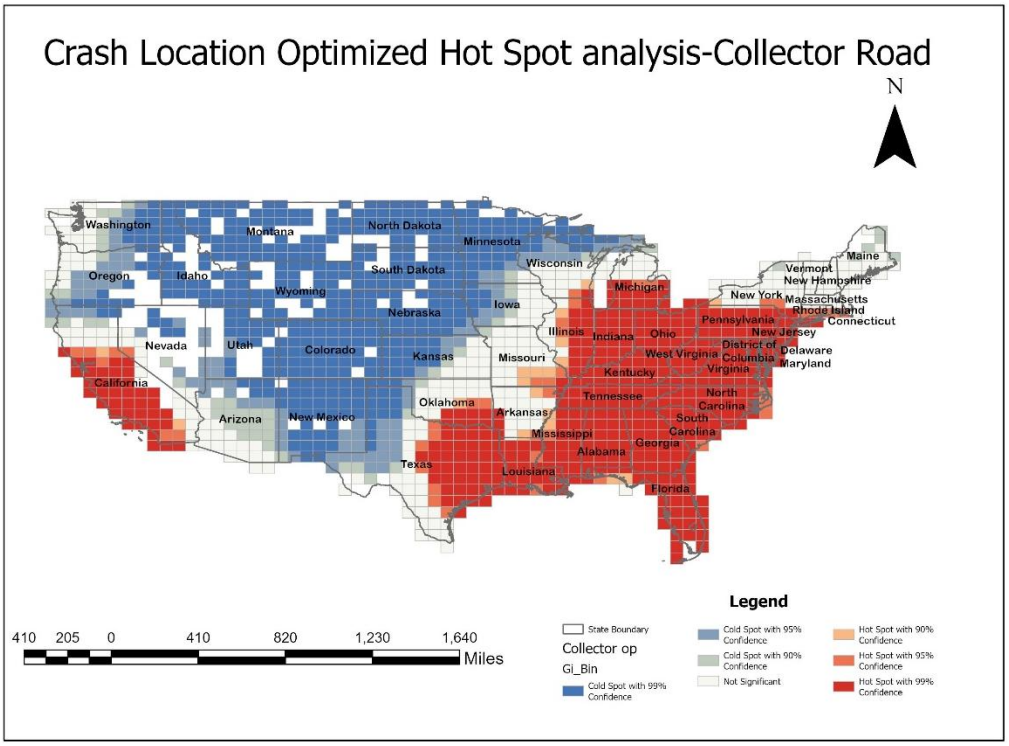


Figure 4.7 Crash location Hot Spot analysis based on Collector Road

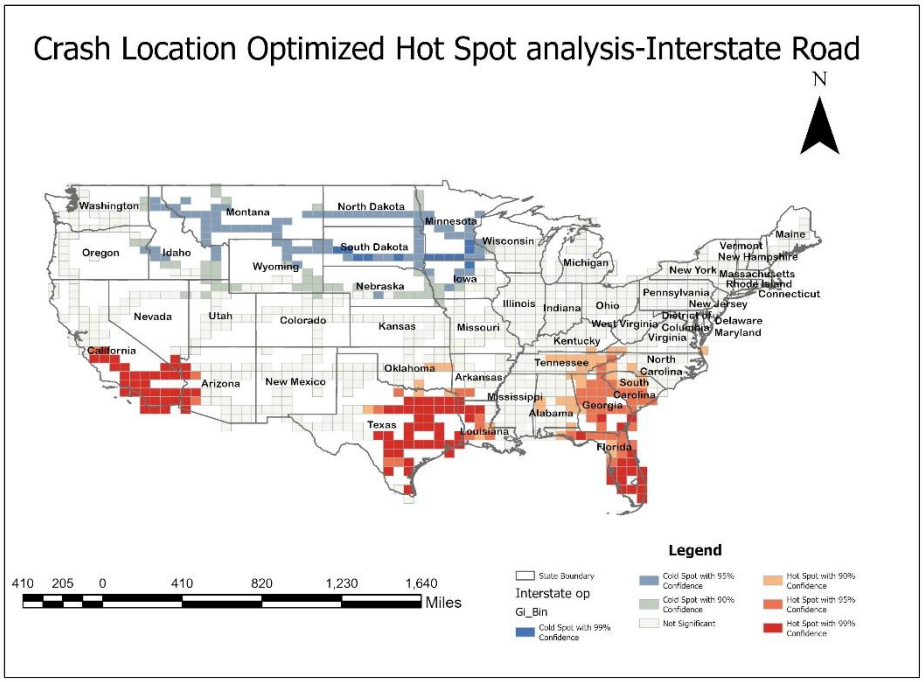


Figure 4.8 Crash location Hot Spot analysis based on Interstate Road

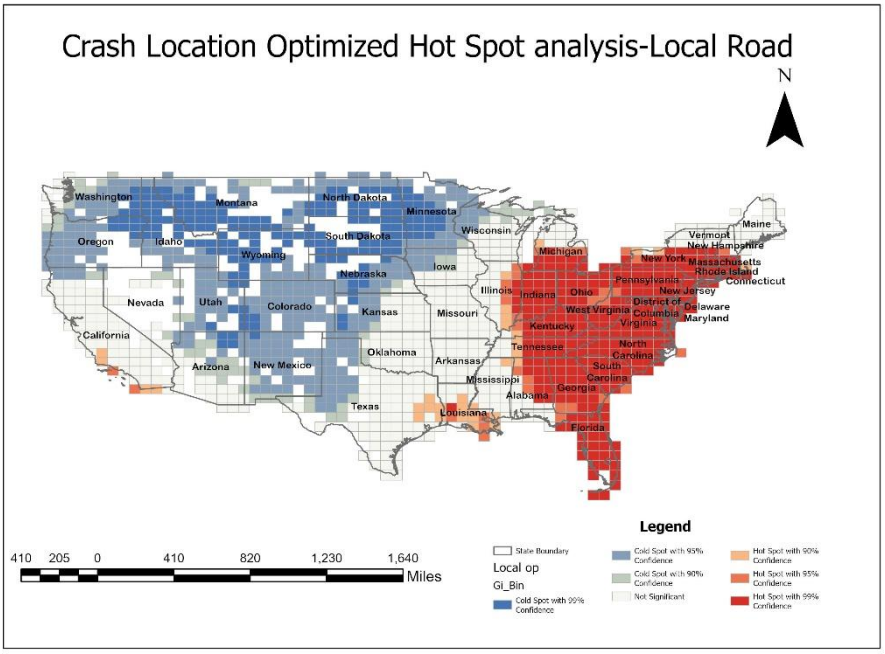


Figure 4.9 Crash location Hot Spot analysis based on Local road

Kentucky, North Carolina, South Carolina, Pennsylvania, New Jersey, Florida, Georgia, Alabama, Louisiana, Virginia showed hot spots with a 99% confidence level. The middle part of U.S. particularly states like Colorado, Kansas, New Mexico, Montana, Idaho, Wyoming, North Dakota and South Dakota showed cold sport with a 99% confidence level. In contrast to collector roads, Texas and California showed cold spots with a 90% confidence level.

4.3.3 Non Spatial Clustering: K-Mode Clustering

Figure 4.10 shows the three diagnostics used to choose k. The K-modes cost falls steadily without a sharp elbow; the silhouette is highest at k = 3 (0.097), and the stability across random starts is also highest at k = 3 (mean adjusted Rand index 0.20). k = 3 was therefore retained. However, silhouette values below 0.25 and an adjusted Rand index of 0.2 indicate weak cluster structure, so the clusters are presented as descriptive profiles of fatal crashes rather than as distinct, stable groups.

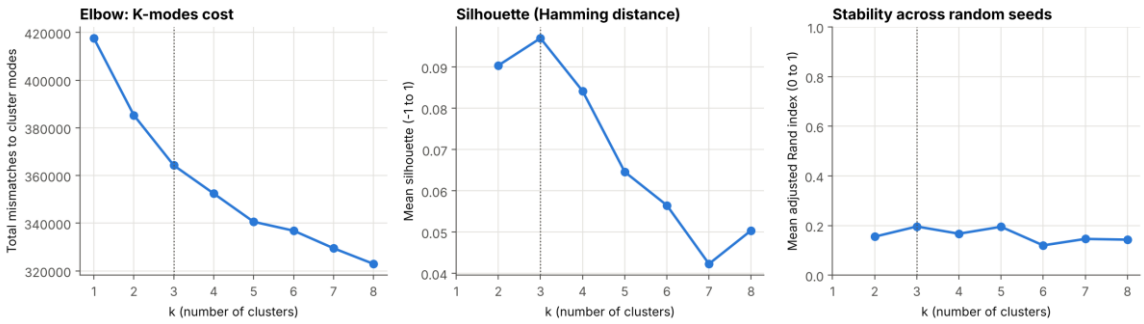


Figure 4.10 K-modes Diagnostics for Choosing k (Cost, Silhouette and Seed Stability)

4.3.4 Cluster Characteristics

K-modes was applied directly to the 18 categorical variables for 306,649 fatalities; unlike k-means it needs no one-hot encoding. Table 4.5 shows, for each variable, the category that is

most over-represented in each cluster, with its share in the cluster and, in brackets, its share among all fatalities.

Table 4.5 Cluster Table

Variable	Cluster 1	Cluster 2	Cluster 3
Size	**137,144 (44.7%)**	**100,433 (32.8%)**	**69,072 (22.5%)**
Road class	Arterial: 67% (all 56%)	Local: 17% (all 12%)	Interstate: 52% (all 13%)
Rural/urban	Urban: 56% (all 55%)	Rural: 49% (all 44%)	Urban: 62% (all 55%)
Race	White: 75% (all 72%)	Black: 17% (all 16%)	Black: 19% (all 16%)
Sex	Female: 30% (all 28%)	Male: 74% (all 71%)	Female: 29% (all 28%)
Person type	Pedestrian: 17% (all 17%)	Motor Vehicle Driver: 68% (all 63%)	Pedestrian: 22% (all 17%)
Age group	45-64: 42% (all 28%)	25-44: 50% (all 33%)	25-44: 44% (all 33%)
Trafficway	Two-Way Not Divided: 79% (all 64%)	Two-Way Not Divided: 78% (all 64%)	Two-Way Divided: 83% (all 33%)
Alignment	Straight: 79% (all 75%)	Curve - Left: 18% (all 12%)	Straight: 83% (all 75%)
Grade	Level: 69% (all 69%)	Grade, Unknown Slope: 11% (all 10%)	Level: 72% (all 69%)
Surface	Dry: 86% (all 84%)	Wet: 14% (all 12%)	Dry: 85% (all 84%)
Intersection	Intersection: 35% (all 24%)	Not an Intersection: 83% (all 75%)	Not an Intersection: 85% (all 75%)
Relation to road	On Roadway: 79% (all 62%)	On Roadside: 57% (all 29%)	On Median: 11% (all 4%)
Light	Daylight: 70% (all 46%)	Dark - Not Lighted: 47% (all 28%)	Dark - Not Lighted: 37% (all 28%)
Weather	Clear: 72% (all 71%)	Cloudy: 14% (all 14%)	Clear: 71% (all 71%)
Lanes	5-6 lanes: 11% (all 9%)	1-2 lanes: 76% (all 68%)	3-4 lanes: 35% (all 21%)
Speed limit	45-55 mph: 54% (all 45%)	45-55 mph: 55% (all 45%)	60+ mph: 65% (all 22%)
Day type	Weekday: 83% (all 67%)	Weekend: 63% (all 33%)	Weekday: 78% (all 67%)
Season	Summer: 39% (all 27%)	Winter: 36% (all 22%)	Fall: 42% (all 27%)

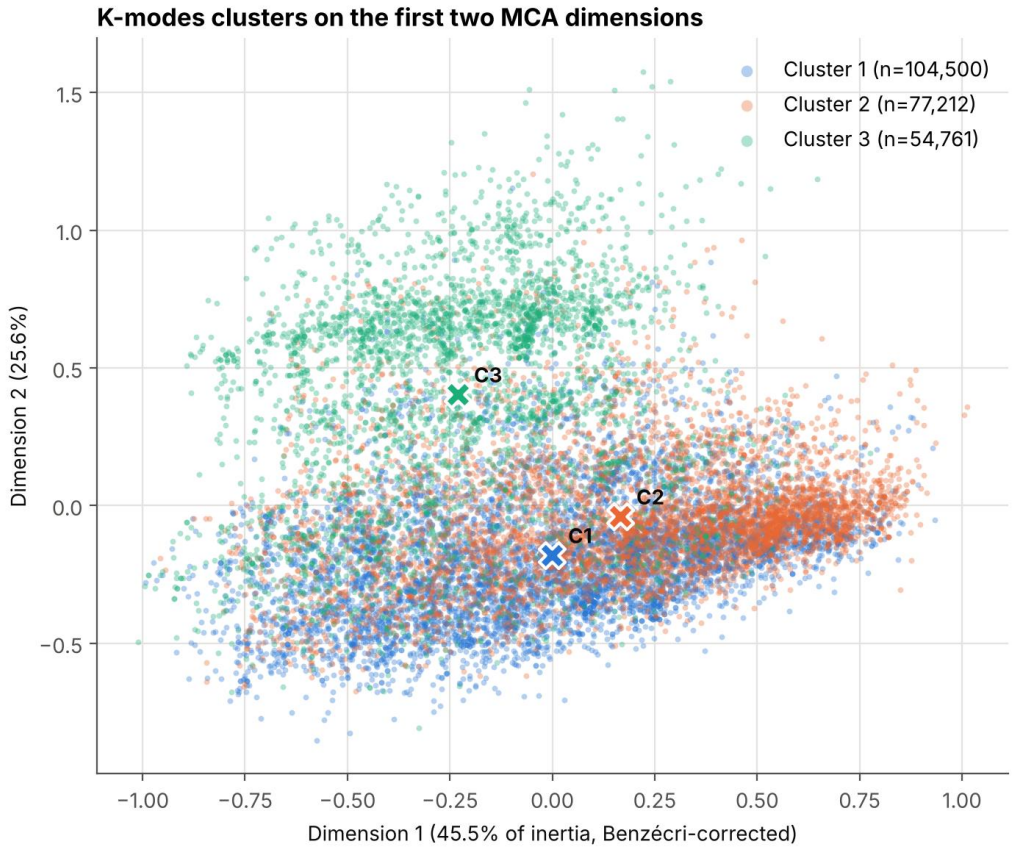
Instead of using principal component analysis, a tool to turn higher dimensional feature into lower one, for visualization, Multiple correspondence analysis is used as all the variables used in this study are categorical. MCA works on the indicator (one-hot) matrix of the categories. It was fitted on the 236,473 fatalities with no unknown value, because with unknown categories included the first dimension only separates records with many unknowns. The first two dimensions explain 4.8% and 4.0% of the total inertia (45.5% and 25.6% after Benzécri's correction; Greenacre, 2017). Dimension 1 separates urban from rural crashes, and dimension 2 picks out high-speed divided highways and interstates (Figure 4.11).

Cluster 1

Cluster 1 (137,144 fatalities, 45%) groups daytime crashes at intersections with older victims. Compared with all fatalities, it has more deaths in daylight (70% vs 46%), at intersections (35% vs 24%), on arterial roads (67% vs 56%) and on weekdays (83% vs 67%), and more victims aged 45–64 (42% vs 28%) and 65 and over (25% vs 18%).

Cluster 2

Cluster 2 (100,433 fatalities, 33%) groups night-time, weekend run-off-road crashes. It has more deaths on the roadside, off the roadway (57% vs 29%), at weekends (63% vs 33%), in the dark without street lighting (47% vs 28%), on left-hand curves (18% vs 12%), on two-lane roads (76% vs 68%) and on local and collector roads, and more victims aged 25–44 (50% vs 33%). Winter is also over-represented (36% vs 22%).



Dim 1 driven by: RUR_URBNAME: Rural, RUR_URBNAME: Urban, LANES_GROUP: 3-4 lanes
Dim 2 driven by: FUNC_SYSNAME: Interstate, SPEED_LIMIT_GROUP: 60+ mph, VTRAFWAYNAME: Two-Way Divided
MCA fitted on fatalities with no Unknown values (236,473); 15,000 randomly sampled points shown; crosses are cluster means.

Figure 4.11 K-modes Clusters on the First Two MCA Dimensions

Cluster 3

Cluster 3 (69,072 fatalities, 23%) groups crashes on high-speed divided highways. Half of its deaths are on interstates (52% vs 13%); most are on divided roads (83% vs 33%) with a speed limit of 60 mph or more (65% vs 22%), with more deaths in the median (11% vs 4%) and in the dark without lighting (37% vs 28%). Pedestrians are slightly over-represented (22% vs 17%).

Season is over-represented differently in each cluster (summer, winter and fall), which is a further sign that the clusters are weakly separated. Compared with the first version of this analysis, the urban pedestrian cluster does not appear in the corrected data: pedestrians are spread across Clusters 1 and 3. The night-time run-off-road profile appears in both versions.

4.4 Concluding remarks

This chapter has discussed detail analysis of the study. Exploratory data analysis gave primary insights of the data which subsequently helped us to conduct spatial and non-spatial clustering method.

Chapter 5

Discussion

5.1 Introduction

This chapter interprets the results of Chapter 4. It summarizes the main findings, compares them with earlier studies, explains what the methods used can and cannot show, and sets out the implications for road safety practice that follow from the findings. Recommendations are limited to measures that respond directly to a result of this study.

5.2 Summary of the Main Findings

Trend. Fatalities rose from 35,484 in 2015 to 43,230 in 2021, an increase of 22%, and fell slightly to 42,514 in 2022. Most of the increase occurred in 2020 and 2021, during the COVID-19 pandemic.

Road class. Arterial roads account for 55.9% of fatalities, collector roads for 18.5%, interstates for 12.9% and local roads for 12.2%.

Associations. Road class is associated mainly with road design and speed. The posted speed limit (Cramér's V = 0.37) and the type of trafficway (0.34) show moderate associations; rural or urban location, pavement type, number of lanes, relation to the roadway and intersection type show weak ones; and the characteristics of the person who died, the weather and the season show negligible ones ($V \leq 0.07$).

Spatial pattern. On all four road classes, fatal crashes are concentrated in the eastern and southern United States and sparse in the mountain west and northern plains.

Profiles. K-modes clustering described three profiles of fatal crashes: daytime intersection crashes with older victims (45% of fatalities), night-time weekend run-off-road crashes on unlit two-lane roads (33%) and crashes on high-speed divided highways (23%). The profiles are weakly separated (silhouette 0.097).

5.3 Comparison with Previous Studies

The finding that road class mainly reflects speed and road design is consistent with reviews of infrastructure-related crash risk factors, which identify alignment, cross-section and roadside design among the road features most strongly related to crash risk (Papadimitriou et al., 2019). It suggests that in FARS the functional class of a road is largely a summary of its design and speed environment rather than an independent factor.

The night-time run-off-road profile matches the roadway departure problem described by the Federal Highway Administration (2021): crashes in which a vehicle leaves its lane, typically on rural two-lane roads, on curves and at night. In this study 37.8% of rural fatalities occurred on the roadside, compared with 21.3% of urban fatalities.

Pedestrians account for 16.8% of fatalities. Three quarters of pedestrian deaths (75.5%) occurred in the dark, 65.3% on arterial roads and 74.5% away from intersections. This agrees with Smart Growth America (2022), which links pedestrian deaths to wide, fast arterial roads. In the cluster analysis, pedestrians do not form a group of their own but are spread between the intersection and high-speed profiles, which suggests that pedestrian crashes are better analysed separately than within a typology of all fatalities.

The over-representation of older victims in daytime intersection crashes agrees with studies showing that older drivers' serious crashes are concentrated at intersections and involve errors in detecting other road users and judging gaps (Caird et al., 2005; Cicchino & McCartt, 2015).

Weather has only a negligible association with road class, and 71% of deaths occurred in clear weather. This does not contradict studies that find a higher crash risk in rain, snow and fog (Tobin et al., 2022): clear weather is by far the most common driving condition, and without exposure data a share of deaths cannot show a difference in risk.

Earlier applications of K-modes to crash data report well-defined groups (Prastya et al., 2021; Tesma, 2020). The diagnostics in this study show that, in a large national dataset, the groups are weakly separated and change with data preparation. This supports reporting stability checks alongside any crash typology.

5.4 What the Methods Can and Cannot Show

Record linkage. In multi-table administrative data, the level at which tables are joined decides what each variable means. Joining race and roadway variables at crash level instead of person and vehicle level would have changed the recorded race of 49,573 people (7,792 of them between the White, Black and Asian groups) and the speed limit of 8,115 people. Person-level linkage, with yearly checks against the official totals, is therefore essential.

Missing-value codes. FARS stores unknown values as numbers (for example 999 for age and 99 for speed limit). Treating them as values would place unknown ages among the oldest victims and unknown speed limits among the highest, so they must be recoded before any grouping.

Significance and effect size. With hundreds of thousands of records every chi-square test is significant. Conclusions in this study rest on Cramér's V, not on p-values.

Cluster stability. The best K-modes solution has a silhouette of 0.097, and solutions from different random starts agree only partly (adjusted Rand index 0.20). The profiles are useful for organising countermeasures but should not be read as distinct populations.

Hot spots. The hot spot maps aggregate crash counts to grid cells, so they show where fatal crashes are concentrated, which largely follows population and traffic volume, not where risk per kilometre travelled is highest. The cell size was chosen separately for each road class, cells without any crash were not analysed, and the grid was not built in an equal-area projection; the maps are therefore descriptive.

5.5 Implications for Road Safety Practice

The measures below respond to the profiles and patterns found in this study. Where possible they are taken from the Federal Highway Administration's list of proven safety countermeasures (Federal Highway Administration, n.d.), which have documented crash reductions.

5.5.1 Night-time Run-off-road Crashes on Two-lane Roads

Profile 2 and the high share of rural deaths on the roadside point to roadway departure. Relevant measures are longitudinal rumble strips and stripes on two-lane roads, enhanced delineation for horizontal curves, wider edge lines, the SafetyEdge pavement-edge treatment and roadside design improvements at curves. Because weekend nights are over-represented, high-visibility enforcement of speeding and impaired driving on weekend nights is a complementary measure; FARS alcohol and drug variables could be added to test this.

5.5.2 Crashes on High-speed Divided Highways

Profile 3 is dominated by interstates and other divided highways with speed limits of 60 mph or more, with deaths in the median over-represented. Median barriers are the main proven countermeasure for cross-median crashes. Lighting at interchanges and fast incident management protect people who are outside their vehicles after a breakdown or an earlier crash.

5.5.3 Daytime Intersection Crashes and Older Road Users

Profile 1 combines intersections, daylight and older victims. Proven measures include roundabouts, reduced left-turn conflict intersections, dedicated left- and right-turn lanes, backplates with retroreflective borders and appropriate yellow change intervals. Design guidance for older road users (legible signs, advance warning of intersections and longer crossing times) addresses the over-representation of victims aged 45 and over.

5.5.4 Pedestrians on Arterial Roads at Night

Most pedestrian deaths occur in the dark, on arterial roads and away from intersections. Measures that match this pattern are lighting, medians and pedestrian refuge islands, crosswalk visibility enhancements, pedestrian hybrid beacons, rectangular rapid flashing beacons and continuous walkways.

5.5.5 Targeting by Location

The hot spot maps identify where fatal crashes are concentrated and can be used to prioritise systemic programmes, such as rumble strips on rural two-lane collector roads in the eastern and southern states. Because the maps are based on counts, individual sites should be selected with rate-based network screening and road safety audits, and local agencies can use local road safety plans for the collector and local roads where hot spots are most widespread.

5.6 Limitations

FARS includes only fatal crashes and contains no exposure data, so this study describes the characteristics of fatal crashes and cannot estimate risk or the effect of a factor on the probability that a crash is fatal. Several variables are police-reported and partly missing. The associations found are descriptive, not causal. National aggregation hides local variation, and the pandemic years 2020–2022 may not represent typical travel. The cluster profiles are weakly separated, and the hot spot maps are based on counts in a grid that was not equal-area.

5.7 Concluding Remarks

The findings point to road design and speed, rather than the characteristics of the people who die, as the main features that distinguish fatal crashes across road classes. The three profiles connect these features to well-established countermeasures, while the methodological lessons show that careful data linkage, treatment of missing values and stability checks are needed before such results are used.

Chapter 6
Conclusion and Further Research

6.1 Conclusion

This study examined fatal traffic crashes in the United States from 2015 to 2022 using all 308,704 fatalities recorded in FARS. It built a person-level dataset from the raw FARS files, validated it against the official yearly totals, and analysed it with exploratory statistics, chi-square tests with effect sizes, K-modes clustering with multiple correspondence analysis, and hot spot analysis by road class.

The first objective was to locate concentrations of fatal crashes on each road class. On arterial, collector, interstate and local roads alike, fatal crashes are concentrated in the eastern and southern states and are sparse in the mountain west and northern plains; because the maps use counts, these concentrations largely follow population and traffic.

The second objective was to group fatalities by their characteristics. K-modes clustering described three profiles: daytime intersection crashes with older victims, night-time weekend run-off-road crashes on unlit two-lane roads, and crashes on high-speed divided highways. The profiles are weakly separated and are best used as a way of organising countermeasures.

The third objective was to test the association between road hierarchy and other variables. All associations are statistically significant, but only speed limit and trafficway type reach a moderate strength; the characteristics of the person who died are negligibly associated with road class.

The fourth objective was to derive recommendations. Each profile corresponds to proven countermeasures: roadway departure treatments for run-off-road crashes, median barriers on high-speed divided highways, intersection improvements for daytime intersection crashes, and lighting and crossing facilities for pedestrians on arterial roads.

In answer to the research question, fatal crashes in the United States group mainly by road environment, time of day and speed rather than by the demographics of the victims, and their spatial concentration on every road class follows the more densely populated eastern and southern states.

6.2 Contribution of the Study

The study does not introduce a new method. Its contributions are practical and methodological. First, it provides a documented and reproducible pipeline from the raw FARS files to person-level, crash-level and spatial datasets, validated against the official totals and published as open-source code. Second, it shows with effect sizes, rather than significance tests, that road class in FARS mainly reflects road design and speed. Third, it offers a typology of fatal crashes together with diagnostics that show how far such a typology can be trusted. Fourth, it documents pitfalls that affect any analysis of FARS: the level at which tables are linked, numeric missing-value codes and invalid coordinates.

6.3 Area of Further Research

Several extensions follow from the limitations of this study. Exposure data, such as vehicle-miles travelled from the Highway Performance Monitoring System or population, would allow crash rates instead of counts to be mapped and modelled, for example with Bayesian spatial models. Network-based hot spot methods, or an equal-area grid with a common cell size, would make the spatial analysis comparable across road classes. Non-fatal crashes from NHTSA's Crash Report Sampling System would allow factors associated with crash severity to be studied. Pedestrian and vehicle-occupant fatalities could be analysed separately, and latent class analysis would give a statistical criterion for the number of groups. Finally, the same workflow could be adapted to cities with limited crash records, such as Dhaka, by combining police data with road features extracted from satellite or drone imagery.

References

Abdel-Rahim, A., & Ellsworth, T. (2021). Effect of pavement condition and geometrics at signalised intersections on casualty crashes. Accident Analysis & Prevention, 149, 105875. https://doi.org/10.1016/j.aap.2021.105875

Adeyemi, O., Paul, R., Delmelle, E., et al. (2023). Road environment characteristics and fatal crash injury during the rush and non-rush hour periods in the U.S.: Model testing and cluster analysis. *Spatial and Spatio-temporal Epidemiology, 44,* 100562. https://doi.org/10.1016/j.sste.2022.100562

Agresti, A. (2013). Categorical data analysis (3rd ed.). Wiley.

Anselin, L., 1995. Local indicators of spatial association—LISA. Geogr. Anal. 27 (2), 93–115. https://doi.org/10.1111/j.1538-4632.1995.tb00338.x.

Bedard, M., Guyatt, G. H., Stones, J. J., & Hirdes, J. P. (2002). The independent contribution of driver, crash, and vehicle characteristics to driver fatalities. *Accident Analysis & Prevention, 34*(6), 717–727.

Bergsma, W. (2013). A bias-correction for Cramér's V and Tschuprow's T. Journal of the Korean Statistical Society, 42(3), 323–328. https://doi.org/10.1016/j.jkss.2012.10.002

Braitman, K. A., Kirley, B. B., Ferguson, S., & Chaudhary, N. K. (2007). Factors leading to older drivers' intersection crashes. *Traffic Injury Prevention, 8*(3), 267–274. DOI: https://doi.org/10.1080/15389580701272346

Caird, J. K., Edwards, C. J., Creaser, J. I., & Horrey, W. J. (2005). Older driver failures of attention at intersections: Using change blindness methods to assess turn decision

accuracy. *Human Factors: The Journal of Human Factors and Ergonomics Society, 47*(2), 235–249.

Chen, C., Zhang, G., & Wang, Y. (2021). Application of machine learning in traffic accident prediction: A review. *Transportation Research Part C: Emerging Technologies, 125*, 103052. https://doi.org/10.1016/j.trc.2021.103052

Chen, H., Cao, L., & Logan, D. B. (2012). Analysis of risk factors affecting the severity of intersection crashes by logistic regression. *Traffic Injury Prevention, 13*(3), 300–307.

Cicchino, J. B., & McCartt, A. T. (2015). Critical older driver errors in a national sample of serious U.S. crashes. *Accident Analysis & Prevention, 80*, 211–219.

Cicchino, J. B. (2015). Why have fatality rates among older drivers declined? The relative contributions of changes in survivability and crash involvement. *Accident Analysis & Prevention, 83*, 67–73.

Federal Highway Administration. (2025). How do weather events impact roads? U.S. Department of Transportation. Retrieved June 13, 2025, from https://ops.fhwa.dot.gov/weather/q1_roadimpact.html

Federal Highway Administration. (2021). Roadway departure strategic plan. U.S. Department of Transportation. https://highways.dot.gov/safety/rwd/resources/roadway-departure-strategic-plan

Federal Highway Administration (FHWA). (2018). *Context Sensitive Solutions in Transportation*. U.S. Department of Transportation.

Federal Highway Administration. (n.d.). Proven safety countermeasures. U.S. Department of Transportation. Retrieved October 23, 2026, from https://highways.dot.gov/safety/proven-safety-countermeasures

Foss, R. D., Martell, C. A., Goodwin, A. H., & O'Brien, N. P. (2011). *Measuring changes in teenage driver crash characteristics during the early months of driving*. AAA Foundation for Traffic Safety.

Gajera, H., & Pulugurtha, S. S. (2024). Fatal crashes involving vehicles with driver warning systems: Identifying risk factors using a correlated random parameters ordered logit modeling approach. *Heliyon*, *10*, e33226. https://doi.org/10.1016/j.heliyon.2024.e33226

Getis, A., Ord, J.K., 2010. The analysis of spatial association by use of distance statistics. Perspectives On Spatial Data Analysis. Springer, pp. 127–145.

Goldberg & Loren Law Firm. (2024). Weather-related accident statistics. Retrieved June 13, 2025, from https://goldbergloren.com/weather-related-accident-statistics/

Greenacre, M. (2017). Correspondence analysis in practice (3rd ed.). Chapman and Hall/CRC.

Guo, M., Yuan, Z., Janson, B., Peng, Y., Yang, Y., & Wang, W. (2021). Older pedestrian traffic crashes severity analysis based on an emerging machine learning XGBoost. *Sustainability*, *13*(2), 926. https://doi.org/10.3390/su13020926

Hamilton-Baillie, B. (2008). Shared space: Reconciling people, places and traffic. Built Environment, 34(2), 161-181.

Ho, G., Scialfa, C. T., Caird, J. K., & Graw, T. (2001). Visual search for traffic signs: The effects of clutter, luminance, and aging. *Human Factors: The Journal of Human Factors and Ergonomics Society, 43*(2), 194–207.

Horgan, D., & Dimitrijević, B. (2019). Frameworks for citizens participation in planning: From conversational to smart tools. Sustainable Cities and Society, 48, 101550. https://doi.org/10.1016/j.scs.2019.101550

Hossein, A. K. M., Golroo, A., & Akhoundzadeh, M. (2024). Smart embedded technologies and materials for enhanced pavement management. Automation in Construction, 168, 105865. https://doi.org/10.1016/j.autcon.2024.105865

Hua, C., & Fan, W. D. (2023). Dynamic speed harmonization for mixed traffic flow on the freeway using deep reinforcement learning. IET Intelligent Transport Systems, 17(11), 2163-2178. https://doi.org/10.1049/itr2.12429

Huang, Z. (1998). Extensions to the k-means algorithm for clustering large data sets with categorical values. Data Mining and Knowledge Discovery, 2(3), 283–304. https://doi.org/10.1023/A:1009769707641

Hubert, L., & Arabie, P. (1985). Comparing partitions. Journal of Classification, 2(1), 193–218. https://doi.org/10.1007/BF01908075

Ji, X., Yue, W., Li, C., Chen, Y., Xue, N., & Sha, Z. (2022). Digital twin empowered model-free prediction of accident-induced congestion in urban road networks. Proceedings of the 2022 IEEE 95th Vehicular Technology Conference (VTC2022-Spring) (pp. 1–6). IEEE. https://doi.org/10.1109/VTC2022-Spring54318.2022.9860491

Jonathan, A.V., Wu, K.F., Donnell, E.T., 2016. A multivariate spatial crash frequency model for identifying sites with promise based on crash types. Accid. Anal. Prevent. 87, 8–16. https://doi.org/10.1016/j.aap.2015.11.006.

Kahane, C. J. (2013). *Injury vulnerability and effectiveness of occupant protection technologies for older occupants and women* (Publication No. DOT HS 811 766). National Highway Traffic Safety Administration.

Kumar, S. S., Vignesh, K. R. N., & Bhuvanesh, V. (2024). Enhancing road safety: Analyzing driver behavior through AI-driven monitoring techniques. In 2024 9th International Conference on Communication and Electronics Systems (ICCES) (pp. 1-8). IEEE. https://doi.org/10.1109/ICCES63552.2024.10859672

Lord, S., Cloutier, M.-S., Garnier, B., Christoforou, Z., 2018. Crossing road intersections in old age—with or without risks? Perceptions of risk and crossing behaviours among the elderly. Transp. Res. Part F 55, 282–296. https://doi.org/10.1016/j. trf.2018.03.005.

Li, X., Liu, J., Zhang, Z., Parrish, A., & Jones, S. (2021). A spatiotemporal analysis of motorcyclist injury severity: Findings from 20 years of crash data from Pennsylvania. *Accident Analysis and Prevention, 151*, 105952. https://doi.org/10.1016/j.aap.2020.105952

Lutz, R. R. (2018). Safe-AR: Reducing risk while augmenting reality. In 2018 IEEE 29th International Symposium on Software Reliability Engineering (ISSRE) (pp. 1-12). IEEE. https://doi.org/10.1109/ISSRE.2018.00018

Mayhew, D. R., Simpson, H. M., & Pak, A. (2003). Changes in collision rates among novice drivers during the first months of driving. *Accident Analysis & Prevention*, *35*(5), 683–691.

McCartt, A. T., & Teoh, E. R. (2015). Tracking progress in teenage driver crash risk in the United States since the advent of graduated driver licensing programs. *Journal of Safety Research*, *53*, 1–9.

McCartt, A. T., Shabanova, V. I., & Leaf, W. A. (2003). Driving experience, crashes, and traffic citations of teenage beginning drivers. *Accident Analysis & Prevention*, *35*(3), 311–320.

National Highway Traffic Safety Administration (NHTSA). (2022). *Traffic Safety Facts Annual Report 2021*. U.S. Department of Transportation. https://www.nhtsa.gov

National Highway Traffic Safety Administration. (2022). Traffic safety facts: 2021 data. U.S. Department of Transportation. https://www.nhtsa.gov/research-data

National Highway Traffic Safety Administration. (2024). Fatality Analysis Reporting System (FARS) analytical user's manual, 1975–2022. U.S. Department of Transportation.

O'Neil, C., & Marsh, F. (2019). Median barriers: Dividing fact from fiction. Transportation Group 2019 Conference Proceedings, 1–9. NZ Transport Agency.

Ord, J.K., Getis, A., 1995. Local spatial autocorrelation statistics: distributional issues and an application. Geogr. Anal. 27 (4), 286–306. https://doi.org/10.1111/j.1538-4632.1995.tb00912.x.

Papadimitriou, E., Filtness, A., Theofilatos, A., Ziakopoulos, A., Quigley, C., & Yannis, G. (2019). Review and ranking of crash risk factors related to road infrastructure. *Accident Analysis and Prevention, 125*, 85–97. https://doi.org/10.1016/j.aap.2019.01.002

Prastya, S. E., Zulfadhilah, M., & Nurhaeni. (2021). Implementation of K-Modes clustering in determining traffic accident patterns. International Journal of Computer Applications, 183(30), 32–37. https://doi.org/10.5120/ijca2021921689

Praticò, F. G., Bosurgi, G., Bruneo, D., Cafiso, S., De Vita, F., Di Graziano, A., Fedele, R., & Pellegrino, O. (2022). Innovative smart road management systems in the urban context: Integrating smart sensors and miniaturized sensing systems. Structural Control and Health Monitoring, 29(5), e3044. https://doi.org/10.1002/stc.3044

Rousseeuw, P. J. (1987). Silhouettes: A graphical aid to the interpretation and validation of cluster analysis. Journal of Computational and Applied Mathematics, 20, 53–65. https://doi.org/10.1016/0377-0427(87)90125-7

Silva, P. (2016). Tactical urbanism: Towards an evolutionary cities' approach? Environment and Planning B: Urban Analytics and City Science, 43(4), 703–722. https://doi.org/10.1177/0265813516657340

Smart Growth America. (2022). Dangerous by design 2022. Transportation for America. https://smartgrowthamerica.org/dangerous-by-design/

Staplin, L., Gish, K.W., Decina, L.E., Lococo, K.H., McKnight, A.S., 1998. Intersection Negotiation Problems of Older Drivers; Volume 1: Final Technical Report (Report No. DOT HS-808-850). National Highway Traffic Safety Administration, Washington, D.C

Sui, D.Z., 2004. Tobler's first law of geography: a big idea for a small world? Ann. Assoc. Am. Geogr. 94 (2), 269–277. https://doi.org/10.1111/j.1467-8306.2004.09402003.

Tesma, G. (2020). Proposing K-mode based methodology for road accident analysis. International Journal of Engineering and Advanced Science and Technology, 66–74. Retrieved from https://www.ijeast.com/papers/66-74,Tesma205,IJEAST.pdf

Tobin, D. M., Reeves, H. D., Gibson, M. N., & Rosenow, A. A. (2022). Motor-vehicle crashes and winter weather: Analysis of fatality risk by precipitation type and visibility conditions in the United States, 2008–2019. Weather, Climate, and Society, 14(3), 833–846. https://doi.org/10.1175/WCAS-D-21-0112.1

Waller, L.A., Gotway, C.A., 2004. Spatial Clustering of Health Events: regional Count Data. Applied Spatial Statistics For Public Health Data. John Wiley & Sons, pp. 236–239. https://doi.org/10.1111/j.1541-0420.2009.01323.

Wilson, E. M., & Lipinski, M. E. (2000). Road safety audits: Applications to urban roads and streets. Transportation Research E-Circular, E-C019, H-1/1 to H-1/8.

Yan, X., Radwan, E., Guo, D., 2007. Effects of major-road vehicle speed and driver age and gender on left-turn gap acceptance. Accid. Anal. Prev. 39, 843–852. doi: http://dx.doi.org/10.1016/j.aap.2006.12.006.

Yan, X., He, J., Zhang, C., Liu, Z., Wang, C., & Qiao, B. (2021). Temporal analysis of crash severities involving male and female drivers: A random parameters approach with heterogeneity in means and variances. *Analytic Methods in Accident Research, 30*, 100161. https://doi.org/10.1016/j.amar.2021.100161

Zhao, Y., Liu, Z., & Zhu, Y. (2020). Spatio-temporal analysis of traffic accidents using geographic information systems: A case study in a metropolitan area. *Journal of Safety Research, 73*, 65–77. https://doi.org/10.1016/j.jsr.2020.03.004

Appendix

All code for this thesis is available at https://github.com/<himal37>/fars-fatal-crash-clustering . The repository keeps the scripts used for the first version of this thesis unchanged in the folder legacy/thesis_version and contains the corrected pipeline used for the results in this version:

01_build_person_dataset.py – merges the FARS person, vehicle, accident and race files at person level and checks the number of fatalities of every year against the crash file.

02_build_crash_points.py – exports crash coordinates by road class for ArcGIS Pro and removes FARS missing-value coordinates.

03_derive_categories.py – creates the age, lanes, speed-limit, day-type and season categories and recodes FARS missing-value codes as Unknown.

04_exploratory_analysis.py – produces the tables and figures of Section 4.2.

05_chi_square_association.py – chi-square tests, bias-corrected Cramér's V and adjusted residuals (Table 4.4).

06_kmodes_clustering.py – K-modes diagnostics, final model and cluster profiles (Figure 4.10, Table 4.5).

07_mca_visualisation.py – multiple correspondence analysis map of the clusters (Figure 4.11).

The file CHANGELOG.md in the repository lists every correction made to the original analysis and the reason for it.

The file docs/statistical_decisions.md in the repository sets out, for each method, its assumptions, why they hold for these data, what would break them and which claims the results can and cannot support.

Arcgis Pro Interface: Hot spot analysis

